

Research paper

Enhancing proactive real-time decision making in manufacturing: A methodological-functional framework

Rocío Mingorance Mingorance ^{a,b,*}, Diego Crespo Pereira ^b, Jone Uribebarria ^a,
Urko Leturiondo ^a

^a Ikerlan Technology Research Center, Basque Research and Technology Alliance (BRTA), Pº. J.M. Arizmendiarrera, 2, Arrasate-Mondragón, 20500, Spain

^b Polytechnic School, University of A Coruña, Ferrol, 15403, Spain

ARTICLE INFO

Keywords:

Digital twin
Decision-making
Optimization
Dynamic scheduling
Machine learning
Manufacturing

ABSTRACT

The manufacturing industry faces major challenges in adapting to automation, driven by market volatility, energy efficiency demands, environmental concerns, and rising competitiveness. Recent advancements in digital technologies and artificial intelligence provide promising tools for manufacturers to tackle operational complexity and enhance sustainability in a demanding market. However, these challenges reveal a critical gap in existing operational frameworks, which often lack mechanisms for proactive, real-time decision-making in response to unforeseen changes in industrial processes.

This paper presents an improved methodological-functional framework that combines physics-based and data-based models to strengthen the alignment between the digital twin and its physical counterpart. It leverages historical data to improve predictions and decision-making, minimizing disturbance impacts and adapting to evolving plant conditions. Unlike traditional models, which do not account for such variations, this framework uses real and synthetic data to generate projections that support proactive real-time decision-making.

Experimental results from a steel industry case study demonstrate that this predictive approach yields an average 11.33% improvement in the objective function, corresponding to an average cost reduction of 133.82 \$/t across the evaluated scenarios. Furthermore, the evaluation of multiple optimization techniques enables customized decision-making under varying timing and operational constraints. The real-time decision-making method attains objective function values up to 1.62% higher than those of metaheuristic approaches, corresponding to cost increases of up to 21.89 \$/t. However, it reduces computation time to 0.27 minutes, compared with up to 25 minutes for metaheuristics, making it suitable for time-constrained, near-optimal applications.

1. Introduction

In an increasingly interconnected world, industries are adopting advanced technologies to improve efficiency and productivity, with the use of Artificial Intelligence (AI) in production processes [1]. Machine Learning (ML) models can learn from normal and faulty operations, predict future scenarios, support proactive planning, and improve decision-making in operational and maintenance scheduling [2,3]. These approaches are widely used to model complex nonlinear relationships [4], delivering gains in operational performance and maintenance cost reduction compared with traditional methods [5]. Their impact is further enhanced by sensors, the Internet of Things, and interconnected databases that enable continuous data analysis, supporting realistic simulations and informed operational and maintenance strategies, and facilitating the transition from static to dynamic production systems [6].

The integration of intelligent systems across manufacturing, from physical assets to plant-level management, drives the transition toward autonomous and cognitive operations [7]. AI algorithms enable the analysis of large volumes of data [8], supporting rapid responses in volatile conditions with fluctuating batch sizes, unpredictable demand, equipment failures, and external disturbances that affect key performance indicators (KPIs) [9].

The Rolling Horizon Procedure (RHP) has been applied to various optimization problems, including the coordination of operations and maintenance in industrial plants [10]. RHP is especially well-suited to address the dynamic complexity and interdependencies that are inherent in sequential optimized decision-making over a planning horizon, an area in which conventional optimization methods encounter significant limitations [11]. These optimization tasks typically involve multiple, sometimes conflicting objectives, such as minimizing operating and

* Corresponding author.

E-mail addresses: r.mingorance@udc.es (R. Mingorance Mingorance), diego.crespo@udc.es (D. Crespo Pereira), JUribebarria@ikerlan.es (J. Uribebarria), uleturiondo@ikerlan.es (U. Leturiondo).

<https://doi.org/10.1016/j.rineng.2026.110033>

Received 26 December 2025; Received in revised form 4 March 2026; Accepted 11 March 2026

Available online 24 March 2026

2590-1230/© 2026 Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Nomenclature

AI	Artificial Intelligence
ABC	Artificial Bee Colony
CCU	Continuous Casting Unit
CO ₂	Carbon Dioxide
DE	Differential evolution
DT	Digital Twin
DTM4SM	Digital Twin Methodology for Smart Manufacturing
DTM4SM-C	Digital Twin Methodological-Functional Framework for Smart Manufacturing with Cognitive Capabilities
EAF	Electric Arc Furnace
ECEA	Enhanced Cross-Entropy Algorithm
EMOSO	Enhanced Multi-Objective Swarm Optimization
FCO ₂	Carbon Dioxide Emission Factor
FLP	Fuzzy Lineal Programming
GA	Genetic Algorithm
GA + RF	RF trained on GA pre-optimized scenarios
HCNSGA-II	Hybrid Chaotic NSGA-II
IFA	Isolation Forest Algorithm
IMOEA	Improved Multi-Objective Evolutionary Algorithm
KPI	Key Performance Indicator
LFU	Ladle Furnace Unit
M2SLH	Multi-Objective Sine Cosine Local Search Heuristic
MAE	Mean Absolute Error
MILP	Mixed-Integer Linear Programming
ML	Machine Learning
MOBWO	Multi-Objective Bee Waggle Optimization
MOHGALS	Multi-Objective Hybrid Genetic Algorithm with Local Search
MOPSO	Multi-Objective PSO
MOWOA	Multi-Objective Whale Optimization Algorithm
MSE	Mean Squared Error
NN	Neural Network
NSGA-II	Non-dominated Sorting Genetic Algorithm II
OF	Objective Function
RF	Random Forest
RHP	Rolling Horizon Procedure
PSO	Particle Swarm Optimization
ROM	Reduced Order Model
SPEA2	Strength Pareto Evolutionary Algorithm 2
SCCSP	Steel-making Continuous Casting Scheduling Problem
XGBoost	eXtreme Gradient Boosting

maintenance costs, reducing process times, and decreasing Carbon Dioxide (CO₂) emissions. Moreover, operation and maintenance planning is a complex and lengthy process due to the high number of possible combinations [12].

Digital Twin (DT) technology provides a high-fidelity virtual representation of physical systems that interacts with real-time data, enabling descriptive, predictive, and prescriptive analytics [13]. Hybrid modeling strategies combining physics-based and data-driven approaches offer a balance between accuracy and computational efficiency, and are increasingly recognized as effective tools for operation and maintenance scheduling in smart manufacturing environments [14,15]. The integration of such hybrid models within a DT framework, together with advanced optimization techniques offers a promising pathway for enhancing decision-making in manufacturing systems.

This paper proposes a Digital Twin Methodological Functional Framework for Smart Manufacturing with cognitive capabilities (DTM4SM-C), extending the original Digital Twin Methodology for Smart Manufacturing (DTM4SM) [16]. The main research gap addressed in this work lies in the limited integration of hybrid modeling, predictive capabilities, and real-time optimization of decision-making within

DT frameworks for real industrial environments, particularly under dynamic disturbances and constraints. The paper is organized as follows: [Section 2](#) reviews related work and further details the identified research gaps; [Section 3](#) presents the DTM4SM-C structure and its decision-making process; [Section 4](#) describes the steel plant use case; [Section 5](#) details the implementation of the framework; [Section 6](#) presents the results obtained under different disturbances and breakages, quantifying the improvements introduced by DTM4SM-C; and [Section 7](#) concludes the paper.

2. Analysis of previous research works, motivation and purpose of the study

This section provides an overview of the research background related to the application of DTs in the steel industry, with special emphasis on the Steel Continuous Casting Scheduling Problem (SCCSP) and decision-support applications. The literature considered in this review has been identified through a structured search in Scopus, Web of Science, and IEEE Xplore, focusing on peer-reviewed contributions published since 2010 that address simulation, prediction, and optimization applications in steel manufacturing. The section emphasizes the use of these techniques to enhance decision-making efficiency. Finally, it summarizes the identified research gap and the motivation underlying this investigation.

2.1. Related works

The existing literature suggests that, despite its potential to improve industrial decision-making, DT approaches still face significant limitations. As noted by Mingorance et al. [16], current DT implementations do not fully integrate simulation, optimization, monitoring, and prediction, restricting their effectiveness as decision-support tools. This shortcoming is particularly critical in complex and variable industrial sectors such as steel manufacturing, where high energy consumption, strict quality requirements, fluctuating electricity prices, and variability in raw material composition hinder process efficiency. These factors often lead to inefficiencies and increased waste, underscoring the need for improved planning and operation strategies.

Since 2010, an increasing number of studies have investigated the SCCSP in dynamic steelmaking environments [17], to optimize production sequencing while minimizing processing time. The production process comprises three main stages: melting in the Electric Arc Furnace (EAF), refining in the Ladle Furnace Unit (LFU), and solidification in the Continuous Casting Unit (CCU). Most existing approaches rely on physics-based models to represent plant behavior and support decision-making. However, they often fail to capture real-time industrial variability, prompting the adoption of hybrid strategies. For example, Long et al. [18] integrated data-driven models to predict job release times, while Zhou et al. [19] employed such models to capture real process variability, providing a more realistic representation of production dynamics.

Mathematical programming has been widely adopted to address the SCCSP. Jiang et al. [17] formulated the problem as a Mixed-Integer Linear Programming (MILP) achieves optimal solutions under deterministic assumptions. Fanti et al. [20] integrated MILP with discrete-event simulation within a DT framework to minimize makespan. However, scalability issues arise as scheduling complexity and uncertainty increase, restricting its use in dynamic environments. Consequently, heuristic and metaheuristic algorithms have been introduced to improve computational tractability. Tan and Liu [21] focused on energy cost reduction through optimized production strategies, Yu et al. [22] addressed delayed job starts via rescheduling mechanisms. Li et al. [23] developed a hybrid Fruit Fly Optimization Algorithm (FFOA) to handle machine failures and Yang et al. [24] introduced an Enhanced Cross-Entropy Algorithm (ECEA) for uncertain processing times. These methods demonstrate improved flexibility and robustness compared to pure MILP

formulations; however, they generally rely on reactive rescheduling rather than anticipatory decision-making.

Metaheuristic-based models further extend this flexibility. Tang et al. [25] proposed an improved Differential Evolution (DE) algorithm for unforeseen system changes, Long et al. [18] employed a Genetic Algorithm (GA) for scheduling optimization, Pan [26] implemented an Artificial Bee Colony (ABC) approach to minimize makespan and waiting times, and Fazel et al. [27] combined Particle Swarm Optimization (PSO) with Fuzzy Linear Programming (FLP). Although metaheuristics provide adaptability, they are typically embedded within reactive frameworks.

Beyond classical objectives such as makespan and waiting time, several studies incorporate energy efficiency and carbon emissions. Xu et al. [28] applied a Multi-Objective Hybrid Genetic Algorithm with Local Search (MOHGALS) to reduce energy losses, while Lian et al. [29] proposed an Improved Multi-Objective Evolutionary Algorithm (IMOE) for energy-aware scheduling. Zhou et al. [19] introduced a fuzzy scheduling model integrating carbon emission reduction through a Multi-Objective Bee Waggles Optimization (MOBWO) algorithm. Although these contributions broaden performance metrics, they largely target isolated units, limiting their applicability to fully integrated, plant-wide DT frameworks.

Multiple studies have addressed DT applications at the unit level. Manojlovic et al. [30] applied ML to improve EAF equipment performance, while Andonovski et al. [31] and Saboohi et al. [32] combined fuzzy logic and optimization for energy estimation and cost reduction. In continuous casting, hybrid DTs and metaheuristic scheduling models have been proposed for predictive maintenance and dynamic planning [33–35]. Additional DT-based solutions have been implemented in refining and cold rolling processes for monitoring and optimization [36,37]. Although these approaches demonstrate the feasibility of DTs at the unit level, they remain fragmented and do not fully integrate decision-making across stages. More recent research has attempted plant-wide integration. Wang et al. [38] proposed a scheduling model encompassing melting, refining, and continuous casting to enhance flexibility and reduce energy costs. However, even these integrated efforts primarily rely on dynamic rescheduling triggered by disturbances, rather than proactive mechanisms capable of anticipating variability and preserving performance under uncertainty.

Table 1 compares the main research efforts on SCCSP, ordered by increasing scheduling complexity. Static scheduling generates a single plan under nominal conditions, assuming fixed load assignments and casting sequences. Dynamic scheduling uses reoptimization in response to disturbances, whereas proactive scheduling anticipates disturbances and produces robust schedules under uncertainty. Static scheduling predominate in the literature, primarily addressing operational constraints and classical objectives such as makespan and waiting times. Dynamic scheduling incorporates metaheuristic models to respond to realized disturbances. Proactive scheduling is comparatively underexplored and is generally supported by hybrid models focused on predicting job release times and temporal variability. The table also enables a comparison in objective functions (OFs) and solution methods. Most studies prioritize processing times, while energy efficiency, emissions, and cost-related metrics receive less attention. Although some metaheuristics have been applied, deterministic mathematical models remain predominant, indicating opportunities for more flexible hybrid approaches.

Most existing studies emphasize the complexity of the SCCSP and the critical role of computational efficiency due to frequent rescheduling and dynamic adjustments. DT applications in this area mainly address time-based planning and scheduling under specific disturbances, including process-time variability and equipment unavailability. These approaches typically rely on physics-based models and optimize classical objectives like makespan, waiting times, and operational costs, while rarely incorporating broader KPIs like CO₂ emissions, total production costs, or resource consumption. Even when metaheuristic al-

gorithms are applied, they usually focus on isolated subproblems and fail to capture plant-level multi-objective optimization, constraints, and interdependencies. These challenges are generally addressed through simulation and optimization to represent industrial process dynamics. However, physics-based models often involve high computational complexity and limited adaptability, restricting their real-time applicability. In steel plants, production and maintenance scheduling is commonly handled using rolling-horizon optimization [41], typically implemented offline. Furthermore, many optimization methods involve trade-offs between robustness and computational effort, limiting their effectiveness in dynamic environments [42].

To address these shortcomings, this article proposes DTM4SM-C. Although the framework is validated through a steelmaking case study, its structure is not sector-specific. The proposed approach is designed for complex industrial environments characterized by multi-stage processes, strong interdependencies, dynamic disturbances, and multi-objective decision-making requirements. By combining hybrid modeling, predictive capabilities, and online optimization within a closed-loop architecture, the framework can be extended to other engineering domains facing similar operational and maintenance challenges.

2.2. Purpose of the study

Motivated by the research gaps identified in the previous section, this study proposes an online closed-loop approach for the real-time adaptation of operating variables and the proactive scheduling of production and maintenance across all process stages, even in the presence of disturbances. To this end, a methodological-functional framework is introduced, combining physical knowledge with data-driven techniques. While physics-based models remain predominant in the literature, this hybrid approach is incorporated into an integrated closed-loop architecture designed to enhance resilience and computational efficiency, as data-driven models can anticipate future scenarios and learn plant behavior under both normal and faulty operating conditions [3,43]. The framework incorporates anomaly detection, future event prediction, and equipment failure probability estimation within a unified structure. Furthermore, it integrates a multi-criteria optimization strategy that balances solution quality and computational response time, supporting effective decision-making across different operational horizons. The main contributions of this paper are summarized as follows:

1. Integrated DT framework: DTM4SM-C is proposed to enable plant-wide integration across all production stages, overcoming the fragmented, unit-level focus of most existing approaches.
2. Hybrid modeling approach: The framework combines physics-based models with data-driven model and ML techniques, achieving improved computational efficiency, predictive accuracy, and resilience under both normal and faulty operating conditions.
3. Online closed-loop and proactive scheduling: An online closed-loop strategy is introduced to adapt operating variables in real time and to support proactive scheduling of production and maintenance, even in the presence of internal and external disturbances.
4. Advanced predictive functionalities: The framework incorporates anomaly detection, future event prediction, and equipment failure probability estimation, enhancing anticipatory decision-making capabilities.
5. Multi-criteria optimization strategy: A flexible decision-making framework with two alternative pathways: a real-time approach delivering fast but approximate solutions, and a quasi-real-time approach providing higher-quality optimal solutions. This enables adaptation across different industrial requirement
6. Industrial applicability to the steel industry: The framework is validated in the context of the SCCSP, demonstrating its effectiveness in improving operational efficiency, resource utilization, and decision-making in highly complex industrial environments.

Table 1

Relevant research on SCCSP, considered disturbances, OF and optimization algorithms. EMOSO: Enhanced Multi-Objective Swarm Optimization; HCNSGA-II: Hybrid Chaotic NSGA-II; M2SLHO: Multi-Objective Local Search Heuristic with Sine and Cosine; MOPSO: Multi-Objective PSO, MOWOA: Multi-Objective Whale Optimization Algorithm; NSGA-II: Non-dominated Sorting Genetic Algorithm II; SPEA2: Strength Pareto Evolutionary Algorithm 2.

Problem	Disturbance/ Uncertainty	Plant Model	OF	Optimization Algorithm	Author
Static scheduling	Operation delay and interruption	Mathematical model	Waiting time	MILP	[17]
Static scheduling	Failures and blocks of machines	Mathematical model	Makespan	MILP	[20]
Static scheduling	Small violations of fuzzy constraints	Mathematical model	Makespan and waiting time	PSO, FLIP	[27]
Static scheduling	–	Mathematical model	Date deviation and energy use	MOHGALS, NSGA-II, SPEA2	[28]
Static scheduling	Uncertain processing time	Mathematical model	Makespan, waiting time and carbon emissions	ECEA, MOPSO, SPEA2, EMOSO, M2SLHO	[24]
Static scheduling	Variable electricity price	Mathematical model	Energy cost	Heuristic	[21]
Static scheduling	–	Mathematical model	Waiting time and makespan	ABC	[26]
Static scheduling	Machine failure and processing time disturbance	Mathematical model	Energy saving	IMOEI NSGA-II, SPEA2, MOHGALS	[29]
Dynamic scheduling	Unforeseen changes in production	Mathematical model	Waiting time	DE	[25]
Dynamic scheduling	Start-time delay	Mathematical model	Waiting time	Heuristic	[39]
Dynamic scheduling	Machine breakdown and processing variation disruptions	Mathematical model	Waiting time, cast-break and system instability	Hybrid Fruit Fly	[23]
Dynamic scheduling	Uncertain release time	Mathematical model	Casting start time, consecutive charges count and system instability	HCNSGA-II, NSGA-II	[34]
Dynamic scheduling	Continuous caster breakdown	Mathematical model	Makespan	GA	[35]
Dynamic scheduling	Uncertain processing and arrival times	Mathematical model	Waiting time and cast-break	DE, PSO	[40]
Proactive scheduling	Uncertain release time	Hybrid approach to predict job release times	Waiting time, cast-break and system instability	GA	[18]
Programación proactiva	Tiempo de liberación incierto	Enfoque híbrido basado en datos para variabilidad real	Tiempo total, tiempo transcurrido y emisiones	NSGA-II, MOBWO, MOWOA	[19]
Proactive scheduling	Machine failures, uncertain processing times, demand increase, variable electricity prices, and charge composition	Hybrid data-driven approach to estimate electrical consumption, temperature, O ₂ concentration, and mold shell strength during casting	Waiting time, casting breakouts, operation and maintenance costs, emissions, water consumption, and demand satisfaction	GA, PSO	This work

3. DTM4SM-C

DTM4SM-C is a methodological-functional framework that enables optimized decision-making in response to disruptions. It comprises a variety of modules that facilitate real-time monitoring, control, prediction, and intelligent decision-making in the manufacturing sector. These modules are configurable, allowing the selection of different modeling approaches for each module according to the amount and quality of available data or the specific characteristics of the problem to be addressed. A connection between the physical plant and the DTM4SM-C is required to maintain the simulation model and data analytics models up to date, as well as transmit the optimized response with the new plant configuration to the control system. Its ability to detect small deviations makes it possible to establish an adaptive control system for key process variables, such as temperature and chemical composition. The incorporation of predictive models, capable of forecasting future scenarios based on current and historical data, together with the optimization of decision-making in real-time, makes it a more comprehensive system than conventional control systems. Fig. 1 shows the schematic structure of the proposed DTM4SM-C. It consists of six main components:

- A simulation model based on discrete-event simulation, system dynamics, or agent-based modeling is required to accurately represent plant behavior. This model can emulate in a detailed and realistic way the interactions between the different components of the system, the material and energy flows, as well as the system responses to scheduled or unexpected events. This is essential to evaluate what-if scenarios, allowing different strategies to be explored without the risks and costs associated with physical testing and without disrupting the operations of the physical system [44]. It also allows the generation of synthetic data under boundary conditions, fault scenarios and operating limits that are difficult to measure in the physical plant.

- A database that includes actual data collected from plant sensors and synthetic data generated using the simulation model. This is crucial for developing and training a reliable Reduced Order Model (ROM) capable of accurately predicting the final state of the plant under normal and abnormal conditions.

When path 1 is selected for the decision-making process, an additional database of optimized scenarios is included. This database enables the model to determine, in real time, the operational variables that need to be adjusted to enhance plant performance in response to disturbances. This database is generated by optimizations based on the ROM and the optimization algorithm.

- Data analytics models detect patterns, predict failure probabilities, and identify anomalies that support optimized decision-making. Anomaly detection models identify abnormal patterns, enabling early fault detection to prevent disruptions to production. Failure probability models calculate the likelihood of equipment failure and assist in prioritizing maintenance tasks, maximizing equipment availability and reducing downtime. Predictive models enable a shift from reactive to proactive strategies, improving the accuracy of scenario anticipation and system availability. To ensure sustained performance and reliability, these models should be periodically evaluated and retrained when significant changes in data distributions are detected.
- A ROM of the manufacturing plant capable of providing KPIs values for any plant configuration and in the event of any potential system disturbance. The ROM runs faster than the simulation model, allowing for streamlined decision-making processes. It leverages both actual plant data and synthetic, ensuring that external influences such as human factors, weather conditions and abrupt changes in plant operation are considered. These unforeseen factors, which are often challenging to predict and model, can have a substantial impact on KPIs. The ROM incorporates these effects, facilitating a more comprehensive and adaptive analysis.

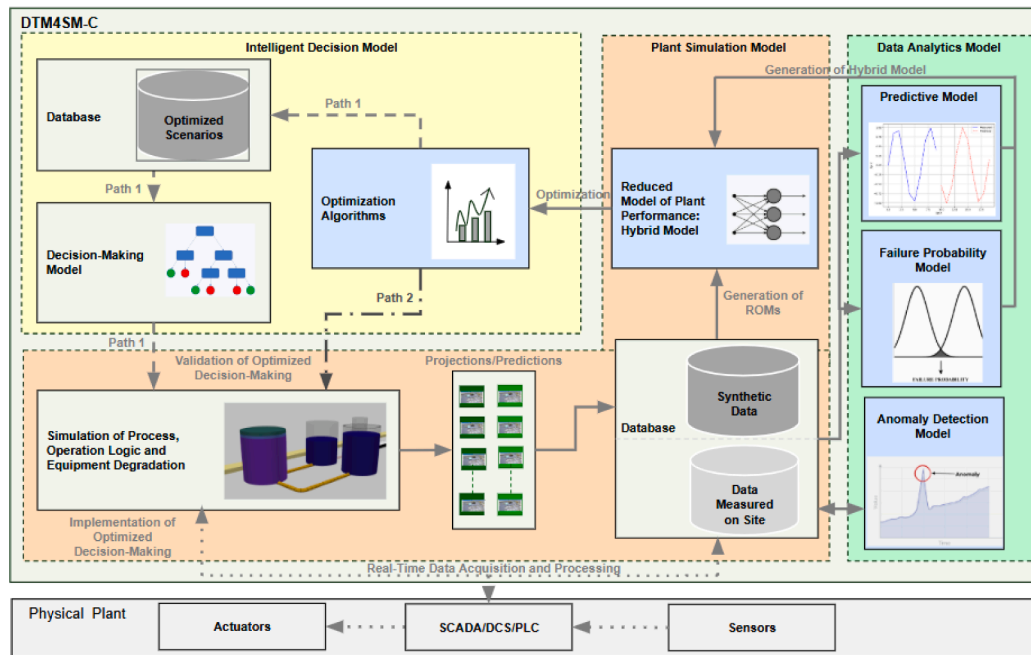


Fig. 1. Schematic structure of the DTM4SM-C. Solid lines denote actions executed by the DTM4SM-C, while dotted lines represent actions performed by external components. Dashed lines indicate the decision-making alternative for Path 1, enabling real-time operation, whereas dash-dot lines correspond to Path 2, supporting near-real-time operation. Newly incorporated models are shown in blue. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

- An algorithm designed to optimize decision-making processes using the ROM model. This algorithm identifies the decision variables of the process that must be modified to effectively mitigate the impact of disturbances. However, the quality of the solution and the speed of convergence can vary significantly depending on the algorithm employed. In scenarios where real-time decision-making within one minute is required, a database containing precomputed optimization results is constructed and used as input for the decision-making model.
- A decision-making framework is designed to evaluate choices and their potential impacts. It is developed by considering the characteristics of the problem as input and the best solution as output. Once trained, the model can predict optimal solutions for new challenges in real-time, thus removing the requirement to execute a time-consuming optimization algorithm. The decision is validated in the simulation model and communicated to the plant control system, enabling real-time adjustments and enhancing the ability of the plant to withstand disturbances.

Algorithm 1 provides the mathematical and procedural representation of the operational cycle of the DTM4SM-C framework.

DTM4SM-C differs from DTM4SM in three key aspects: the hybrid modeling strategy used to construct the ROM, the integration of data analytics models, and the incorporation of enhanced optimization and decision-making mechanisms. In DTM4SM-C, the ROM is developed using a hybrid approach that combines physics-based models to represent internal plant processes with data-driven models that capture external influences and complex behaviors not easily described by analytical equations. In addition, DTM4SM-C integrates data analytics models to enhance the characterization of the current and future states of the plant. These include predictive models, anomaly detection models, and equipment failure probability estimators, enabling the DT to autonomously identify unexpected behavior, anticipate failures, and proactively trigger prediction and optimization processes without relying on external supervision systems. By contrast, DTM4SM relies on anomaly detection performed externally by existing plant control systems, such as SCADA

Table 2
Comparison of the two ROMs.

Characteristics	ROM in DTM4SM-C	ROM in DTM4SM
Physical equations	Yes	No
Simulation data	Yes	Yes
Measured plant data	Yes	Yes
Data analytics models	Yes	No

or PLCs, and its ROM is constructed from data generated by the simulation model and plant measurements. [Table 2](#) summarizes the main differences between the ROM implementations in both frameworks.

In addition, DTM4SM incorporates an evolutionary algorithm to optimize the decision-making process. However, due to their high computational cost and slow convergence, these algorithms are not suitable for online execution. To address this limitation, the approach includes a database of pre-optimized scenarios and applies a supervised learning algorithm to accelerate decision-making. Although this reduces execution time, it may compromise solution quality. To adapt solution quality and computational time to the requirements of each industry, the DTM4SM-C introduces an alternative decision path that evaluates different optimization algorithms to select the one that best meets the time and performance requirements of each application.

The implementation of the proposed framework follows a structured and systematic procedure that begins with the formal definition of its objectives and operational scope. This phase establishes the intended functionalities of the DT in relation to the physical system. Concurrently, system boundaries and relevant subsystems are defined to ensure an adequate level of model fidelity, enabling an accurate representation of the physical structure, dynamic behavior, and operational constraints of the real system.

Subsequently, the critical variables and parameters influencing the KPIs are identified through the analysis of industrial data, technical specifications, scientific literature, and sensitivity studies. Model assumptions and constraints are then formulated to strike a balance between predictive accuracy and computational tractability, which is

Algorithm 1 DTM4SM-C operational cycle.

```

1: while real-time operation at time  $t$  do
2:   1. Data acquisition:
3:    $x_t \leftarrow \text{decision\_variables}()$   $\triangleright$  Current values of decision variables
4:    $y_t \leftarrow \text{sensor\_measurements}()$   $\triangleright$  Sensor readings
5:    $u_t \leftarrow \text{disturbance}()$   $\triangleright$  External disturbances
6:   2. Anomaly detection model:
7:    $\text{alerta}_t \leftarrow A(y_t)$   $\triangleright$  Detect anomalies using model  $A$ 
8:    $\text{SCADA.send}(\text{alerta}_t)$   $\triangleright$  Send detected anomaly to SCADA
9:   3. Simulation model  $\triangleright$  Estimate KPIs under different configurations
10:   $k_t^{\text{sim}} \leftarrow \text{Simulator.f}(x_t, u_t)$   $\triangleright$  Simulated KPIs
11:   $\text{D1.add}(x_t, y_t, u_t, k_t^{\text{sim}})$   $\triangleright$  Store data in database D1
12:  4. Data analytics models:
13:   $e_t^{\text{pred}} \leftarrow P(x_t, y_t)$   $\triangleright$  Event prediction using model  $P$ 
14:   $r_t \leftarrow R(x_t, y_t)$   $\triangleright$  Failure probability / risk using model  $R$ 
15:  5. ROM:
16:   $k_t^{\text{rom}} \leftarrow \text{ROM}(x_t, y_t, u_t, \text{alerta}_t, e_t^{\text{pred}}, r_t)$   $\triangleright$  Compute KPIs using the ROM
17:  6. Optimization:
18:   $x_{\text{opt}} \leftarrow \text{Optimizer}()$   $\triangleright$  Optimized decision variables
19:   $\text{objective} = J(k_t^{\text{rom}})$   $\triangleright$  OF to minimize/maximize
20:  )
21:  Decision path 1:
22:   $\text{D2.add}(x_t, u_t, y_t, x_{\text{opt}})$   $\triangleright$  Store optimized scenario in database D2
23:   $x_{\text{opt}}^{\text{dec}} \leftarrow \text{Decision\_model}(x_t, u_t, y_t)$   $\triangleright$  Execute decision-making model
24:  7. Validation using simulation model:
25:   $k_{\text{opt}}^{\text{sim}} \leftarrow \text{Simulator.f}(x_{\text{opt}}, u_t)$   $\triangleright$  Compute optimized KPIs
26:  8. Send optimized action to SCADA:
27:   $\text{SCADA.send}(x_{\text{opt}})$   $\triangleright$  Send optimal solution to SCADA
28:  9. ML model update:
29:  if  $(x_t, u_t) \notin D_1 \wedge (x_t, u_t) \notin D_2$  then
30:     $\text{ROM.train}(D1)$   $\triangleright$  Update ROM
31:     $\text{Decision\_model.train}(D2)$   $\triangleright$  Update decision-making model
32:  end if
33: end while

```

essential for real-time decision-making. Based on these considerations, an appropriate modeling paradigm is selected to capture system dynamics and anticipate variations under different operating conditions. During the validation and documentation of the framework, outputs are rigorously compared against real plant data under nominal and extreme operating scenarios to assess accuracy, robustness, and predictive performance. All modeling decisions, including parameter selection, assumptions, constraints, and validation outcomes, are documented to ensure transparency, reproducibility, and reliable deployment in industrial environments.

Ensuring interoperability with existing SCADA, DCS, and PLC systems is also essential to achieve seamless integration and real-time responsiveness. The proposed framework is integrated through standardized data exchange interfaces that enable access to real-time operational information and allow the synchronization of model-generated recommendations with control and management platforms. For real-time execution, the system requires continuous data streams of main process variables and equipment condition indicators. These data streams must be consistent and sufficiently frequent to support the dynamic updating of predictive models, anomaly detection, and the recalibration of operational strategies, ensuring alignment with the current state of the process. Although the proposed framework does not explicitly address communication protocols, it remains compatible with existing industrial process control infrastructures through the use of standard communication protocols, avoiding the need to replace operational components [45].

4. Definition of the use case

The metals industry is a key component of the global economy, converting metallic minerals into products used across many industrial sectors. Steel production involves complex and interconnected processes such as melting, refining, continuous casting, and thermomechanical treatment [46]. Steel mill operation is highly dynamic and requires integrated strategies to monitor processes, detect disturbances, and rapidly adjust operating conditions. Effective responses are essential to minimize downtime, optimize resource use, and reduce the impact of disturbances on KPIs. Disturbances can arise from both internal and external sources and affect multiple stages of production. Variations in raw material composition and operating parameters during melting can reduce energy efficiency and product quality. Similarly, disruptions in continuous casting may cause defects such as cracks or inclusions, leading to yield losses and higher reprocessing costs.

Therefore, the steel industry is selected as a use case to implement the proposed methodological-functional framework by utilizing the minute-level data provided by [47]. This data includes detailed information on material charging, transformer operation stages with electricity consumption, temperature and oxidation measurements, cumulative and flow rates of gas, oxygen, and injected carbon, as well as chemical composition measurements taken at different stages of the process during the period 2015-2018. It also covers material additions during tapping, the initial chemical analysis in the LFU, and any additions made to the LFU. For the CCU, data provided by [48] for the period from January to August 2020 has been used. The data were collected from a real production facility during CCU operation, where the automatic control system recorded casting parameters, including workpiece weight, steel type, cast-in row, number of crystallizers, workpiece slice geometry, resistance, water consumption, and temperature. After removing the liner from the mold for inspection, the data were retrieved from the SCADA system.

4.1. Process plant description

The most important processes in the steel industry are melting, refining, and continuous casting. During these stages, multiple units can operate in parallel, ensuring a continuous production flow and improved efficiency. Since the material remains in a liquid state at extremely high temperatures, precise coordination is essential to minimize energy consumption, heat loss, and production delays, while ensuring smooth transitions between interconnected processes. Fig. 2 illustrates the workflow of these processes.

4.1.1. Fusion

In steel production, the EAF process is the main method for melting recycled steel, iron scrap, pig iron, and direct reduced iron into carbon steel [49]. The EAF is a batch process involving charging, preheating, melting, and tapping [50]. Melting begins at low voltage with a short arc, then gradually increases. The EAF melts the metal and slagging materials, reaching the casting temperature while offsetting heat losses. Selecting the wrong scrap during melting can lead to electrode breakage or molten steel spillage. [51]. Arc power and length are controlled by choosing the appropriate transformer stage. The oxidation period begins after the charge is fully melted and the bath is flat. During this time, a sample is taken to monitor the chemical composition and temperature with an automatic unit. The gas-oxygen module heats and melts the bulk charge while generating slag froth during the refining process. At the end of the oxidation period, the decision to tap the heat is based on the carbon content measured by the device.

4.1.2. Refining

The LFU is a metallurgical unit designed for treating steel outside the primary furnace. It facilitates key processes such as desulfurization, chemical composition adjustment, and sampling. The LFU consists of a

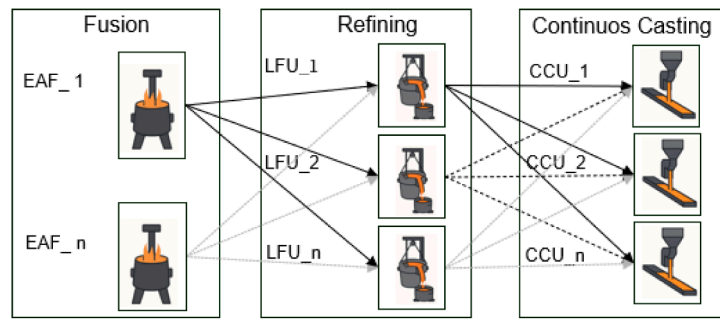


Fig. 2. Workflow of the primary processes of the plant, where multiple units within the same stage operate in parallel and transfer materials to the best option in the subsequent process.

large metallic ladle, lined with refractory bricks to withstand high temperatures. Molten steel is transferred into the LFU from a steel ladle and heated by graphite electrodes through the lid. Alloying uses ferroalloys introduced through a bulk material supply system and wire materials delivered via a tubular apparatus. The chemical composition and temperature of the molten steel are measured using a bottom purge device in the steel ladle each time the molten steel is introduced into the LFU. If needed, the melt is heated for a specified duration. Alloying materials are then added, and the melt is purged with gas to facilitate stirring. This cycle is repeated until the desired chemical composition and melting temperature are achieved. Once the treatment in the LFU is complete, the molten steel is directed to a continuous casting facility, where it solidifies into metallurgical slabs for further processing.

4.1.3. Continuous casting

A CCU is a machine that transforms liquid steel into solid billets of a specified cross-section. When casting begins, the nozzle at the bottom of the ladle opens, allowing steel to flow at a controlled rate into the tundish and through a submerged entry nozzle into one or more molds [52]. The downward speed of the solidified bars is approximately 10% lower than the speed of the ingot mold. [53]. The mold sleeve is the most critical and fastest-wearing component of the CCU. It is a water-cooled copper tube, where molten metal crystallizes upon contact with the sleeve walls, forming the solid layer of the ingot. The primary production issue during jacket operation is surface defects on the copper tube, which distort its inner cavity and affect ingot quality, potentially leading to shape defects and cracks. Additionally, the lifespan of the copper sleeves used in production molds is diminished, often due to variations in operating parameters such as the temperature of the incoming molten metal and the cooling water.

4.2. Problem description

The operation of a steelmaking plant constitutes a highly complex and dynamic problem in which multiple production stages and resources interact continuously under diverse constraints. Ensuring stable and efficient operation requires the capability to monitor the system state, diagnose deviations, and respond promptly to disturbances to minimize downtime, preserve product quality, and maintain overall process performance. Production and maintenance scheduling in steel plants must simultaneously address multiple, often conflicting, objectives, including cost and delay minimization, energy consumption reduction, product quality assurance, and equipment availability, all under strict operational and environmental constraints.

Steelmaking processes are subject to both internal and external disturbances. In the melting stage, variations in the chemical composition of raw materials lead to deviations in metallurgical reactions, directly affecting energy efficiency and the quality of the produced steel. These variations hinder the selection of consistent operating parameters and propagate their effects to downstream processes. In addition, anomalies

in critical operating variables, such as temperature or oxygen concentration, may compromise compliance with quality specifications. In the CCU, process interruptions or inadequate operational adjustments can result in material defects, such as cracks or inclusions, thereby reducing process yield and increasing reprocessing costs. Furthermore, mold lifetime exhibits significant variability, with typical values ranging from 100 to 300 cycles [53]. This variability poses a substantial challenge to operational stability and to production and maintenance planning, as it requires anticipatory replacements to avoid unexpected failures or degradation in product quality.

As a result, assessing the current state of the plant and determining corrective actions after a disturbance represents a highly complex problem. Disturbances may propagate across different process stages, affecting key performance indicators related to productivity, quality, energy consumption, and operational costs. In this context, continuous KPI monitoring is essential to evaluate plant performance, to enable the early detection of anomalies and to support timely and effective responses in dynamic operating environments. When a disturbance is identified, its impact on the KPIs must be quantified, and the decision variables that require adjustment must be determined to restore acceptable operating conditions. This task is particularly challenging in large-scale industrial systems, where subsystem interactions, nonlinear behaviors, and inherent variability complicate accurate anticipation of the system response.

5. Application of DTM4SM-C to the use case

The steel plant model is developed by combining data-driven and physics-based models. The overall modeling framework can therefore be characterized as hybrid, since it integrates first-principles formulations with data-driven methods [54]. The system is simulated at a macroscopic level, focusing on melting, refining, and continuous casting units. The model assumes constant loads for the EAF and LFU between optimization steps, while their processing times are considered uncertain due to varying operational factors. The LFU is represented by two parallel machines. Three types of end products with different shapes and weights are considered. CCU consists of three production lines, each dedicated to a specific product type and composed of eight sleeves. Ladles are assigned according to alloy requirements. Operational parameters can be adjusted to meet production demand efficiently within time constraints while minimizing costs.

5.1. KPIs estimation

To provide a clearer depiction of the production process, the current scheduling of steel production employs discrete time representation methods. Compared with continuous time, discrete time has been proven to be more appropriate for industrial batch production [38]. The parameters and variables necessary to accurately model the process are included in Table 3.

Table 3

Main information for the representation of the physics-based model.

Notations	
Inputs:	
A_t	Electricity grid price (\$/kWh) at time t , $A \in \mathbb{R}$
$B_{r,t}$	Total number of pieces demanded of type r at time t , $B_r \in \mathbb{N}$
C_e	Equipment power consumption (kWh/t), $C_e \in \mathbb{R}$
D_t	Disturbance at time t , $D_t \in A, B, C$
e	Equipment identification, ($e = 1, 2, \dots, 6$)
F_e	Equipment water consumption factor, $F_e \in \mathbb{R}$
H	Period of time between t and T (h), $H \in \mathbb{N}$
J	Completion time of the last piece (h), $J \in T$
M_e	Required maintenance time of equipment e (h), $M_e \in \mathbb{N}$
P	Cost of raw materials (\$/t), $P \in \mathbb{N}$
r	Type of manufactured part, ($r = 1, 2, 3, 4$)
T	Total planning horizon (h), $T \in \mathbb{N}$
$t_{A,B,C}$	Time in which D occurs (h), $t \in T$
U	Amount of raw materials (t/t of steel), $U \in \mathbb{N}$
$\alpha, \beta, \gamma, \zeta$	Linear weighting weights $\alpha, \beta, \gamma, \zeta \in \mathbb{R}$
η, θ, ι	Penalty factors, $\eta, \theta, \iota \in \mathbb{R}$
Decision variables:	
K_e	Equipment load (t), $K_e \in \mathbb{N}$
L_e	Equipment operation mode ($L = 0, 1, 2, 3$)
$O_{e,r}$	Process time (min), $O_{e,r} \in \mathbb{N}$
Outputs:	
G_t	Tons of parts manufactured at time t , $G \in \mathbb{N}$
$N_{r,t}$	Number of parts manufactured of type r at time t , $N \in \mathbb{N}$
S_e	Idle time of equipment e (h), $S_e \in \mathbb{R}$
KPIs:	
$Cost$	Total cost (\$), $Cost \in \mathbb{R}$
CO_2	Emissions of CO_2 (t/t), $CO_2 \in \mathbb{R}$
CpT	Cost per ton (\$/t), $CpT \in \mathbb{R}$
$DemandC$	Compliance with demand (%), $DemandC \in \mathbb{R}$
PpM	Number of pieces produced per minute, $PpM \in \mathbb{R}$

Table 4

Summary of initial conditions and operation and maintenance parameters.

Parameters	Value	Initial Conditions	Value
Weight piece 1 (kg)	20	Demanded piece 1	50,000
Weight piece 2 (kg)	10	Demanded piece 2	40,000
Weight piece 3 (kg)	15	Demanded piece 3	60,000
EAF water (m^3 /t of steel)	3.3	C_1 (kWh/t)	350–700
LFU water (m^3 /t of steel)	3.6	$C_{2,3}$ (kWh/t)	350–600
CCU water (m^3 /t of steel)	2.5	C_{4-6} (kWh/t)	5–15
EAF FCO_2 (t/t of steel)	0.08	M_e (h)	6
LFU FCO_2 (t/t of steel)	0.002	Duration of operation stops (h)	4
CCU FCO_2 (t/t of steel)	0.005	Initial resistance of the sleeve (t)	5000

In Table 4, the demand and weight for each product type considered in this use case are presented, together with the corresponding operational and maintenance parameters. The Carbon Dioxide Emissions Factor (FCO_2), as well as water and electricity consumption per ton of steel produced at each process stage, are taken from [55,56], and [57]. The estimated electricity price (kWh) for the modeled week is based on data from Red Eléctrica, the operator of the Spanish electricity system [58], with hourly variations following the pattern shown in Fig. 3.

The KPIs have been selected for their relevance in evaluating process performance and their importance in supporting decision-making and are defined as follows:

- **Cost:** This KPI reflects the total costs associated with steel production over a specified time frame. It encompasses expenditures related to raw materials, as well as the consumption of electricity, water, among other expenses.

$$Cost = U \cdot G_T \cdot P + \sum_{i=1}^T \left(A_t \cdot \sum_{e=1}^6 PCE_{i,t} \right) \quad (1)$$

- **CO_2 :** This KPI measures the CO_2 emissions generated per ton of steel produced, allowing the steel plant to assess its environmental per-

formance and ensure compliance with current regulations.

$$CO_2 = G_t \cdot \sum_{i=1}^e FCO_{2,i} \quad (2)$$

- **DemandC:** This KPI measures the alignment between the quantity of manufactured parts of each type and the corresponding demand. It helps determine whether system recalibration is needed in cases of underproduction or overproduction.

$$DemandC = \frac{N_{r,t}}{B_r} \cdot 100 \quad (3)$$

- **CpT:** This KPI assesses the expense associated with manufacturing one ton of steel. It is determined by dividing the overall production expenses by the total steel output within a defined time frame. This metric is essential to determine the economic efficiency of the plant and identify deviations.

$$CpT = \frac{Cost}{G_T} \quad (4)$$

- **PpM:** This KPI measures the production rate of a manufacturing process by calculating the number of pieces produced in one minute. It is essential for evaluating the efficiency and productivity of production operations.

$$PpM = \frac{\sum_{r=1}^n N_{r,t}}{J \cdot 60} \quad (5)$$

In order to mitigate the impact of disturbances on KPIs during a production period in which demand fulfillment is essential and considering a production system comprising multiple pieces of equipment and various products, within a defined time horizon T , it is imperative that the system accurately determines the new value of the decision variables. This optimization process must adhere to the constraints established in Eqs. 6 and 7, ensuring that both production and maintenance objectives are effectively met.

$$\sum_r B_r \leq N \quad t \leq T \quad (6)$$

$$S_e \geq M_e \quad t \leq T \quad (7)$$

The OF aims to minimize production costs, surplus, CO_2 emissions, and water consumption while satisfying constraints. A scalarization approach is employed to transform the multi-objective problem into a single-objective formulation. The weights α , β , γ , and ζ are derived using a combination of range-based normalization and the CRITIC (Criteria Importance Through Intercriteria Correlation) method, which objectively assesses the relative importance of each criterion based on its variability and independence [59]. The resulting weights are $\alpha = 6.50 \times 10^{-3}$, $\beta = 6.21 \times 10^{-1}$, $\gamma = 3.48 \times 10^{-1}$, and $\zeta = 7.29 \times 10^{-2}$. To prioritize weekday operation, penalty factors of 0.03 and 0.05 are applied to Saturdays and Sundays, respectively. Eq. (8) presents the optimization problem, formulated as the minimization OF.

$$\begin{aligned} \min_{\mathbf{X}} F(\mathbf{X}) = & \alpha CpT(\mathbf{X}) + \beta CO_2(\mathbf{X}) + \gamma \sum_{i=1}^e Water_i(\mathbf{X}) \\ & + \zeta (100 - DemandC(\mathbf{X}))^2 + \begin{cases} \eta, & \text{if } J(\mathbf{X}) \leq 120, \\ \theta(J(\mathbf{X}) - 120), & \text{if } 120 < J(\mathbf{X}) \leq 144, \\ \iota(J(\mathbf{X}) - 144), & \text{if } J(\mathbf{X}) > 144. \end{cases} \end{aligned} \quad (8)$$

where:

$$\mathbf{X} = \{K_e, O_{e,r}, L_e\} \quad (9)$$

At the beginning of the time horizon, the system is initialized in a previously optimized state for all T . This optimization can be performed offline before T begins, based on the system boundary conditions. When a disturbance occurs, its impact on the KPIs over T must be evaluated

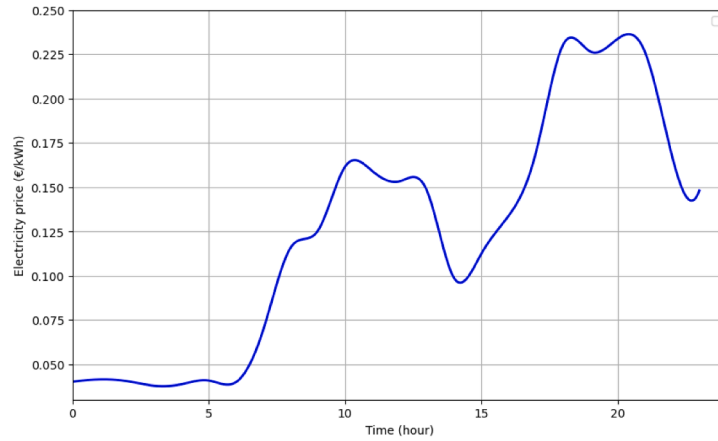


Fig. 3. Hourly electricity price evolution for a single day.

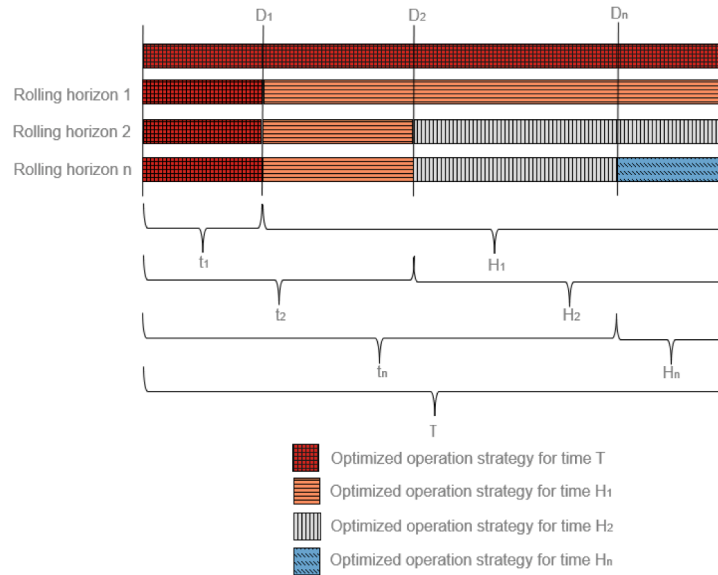


Fig. 4. Rolling horizon planning: Following each disturbance, a newly optimized operational strategy is determined and implemented for the remaining duration of the planning horizon H .

to define corrective actions that restore optimal operation. This continuous optimization process spans both production planning and equipment maintenance across the planning horizon H .

This optimization process entails adjusting decision variables while accounting for the planning period H , the remaining quantity of parts to be manufactured, fluctuations in electricity prices over H , and the scheduled downtime of each piece of equipment within H . Fig. 4 illustrates the iterative resolution of a time-dependent model, which is recalculated whenever a disturbance occurs that impacts KPIs negatively. Additionally, the optimized interval is progressively advanced with each new solution. The figure depicts how, upon the occurrence of disturbances D_1 , D_2 , D_n at the specific times t_1 , t_2 , t_n , the operational strategy is accordingly adjusted, transitioning to a newly optimized configuration for the remaining duration of the respective time horizons H_1 , H_2 , H_n .

5.2. Data-driven model

Data-driven models are used to simulate the behavior of industrial plants by capturing relationships between variables and accounting for factors that are difficult to model explicitly. They enable pattern recognition, anomaly detection, and prediction of failures. These models support proactive problem resolution, facilitate continuous improvement

in plant operations and provide a robust basis for informed decision-making.

To enhance system performance and improve process understanding, four models have been developed for the EAF. These include two anomaly detection models, along with two predictive models: 1 aimed at estimating the chemical composition prior to the refining phase, and another focused on predicting power consumption. In addition, a failure probability model has been designed to evaluate the resistance of the CCU sleeve. The development of these models is based on data collected between January 1, 2015, and July 31, 2018, obtained from the datasets reported in [47] and [48]. The breadth and continuity of these data provide a solid foundation for analysis, enabling the identification of relevant patterns and trends across the process.

5.2.1. Anomaly detection model

Anomaly detection facilitates timely interventions, reducing downtime and ensuring operational stability. This study develops two unsupervised models for key EAF variables: temperature, critical for steel quality and refractory protection, and oxygen concentration, essential for impurity oxidation, combustion efficiency, and operational safety. Given the lack of fully labeled anomaly datasets in industrial practice, an unsupervised approach has been adopted. Both models use the Isolation Forest Algorithm (IFA), a robust method that isolates outliers through

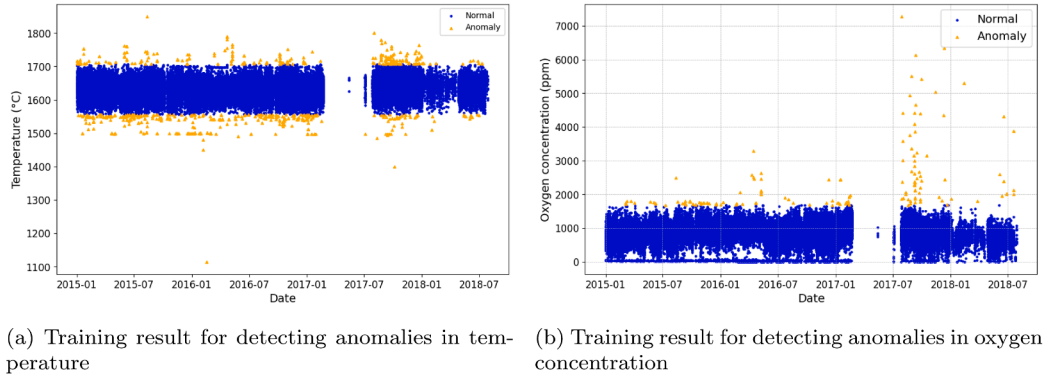


Fig. 5. Anomaly detection using real EAF data with the IFA.

Table 5
Settings for anomaly detection models.

Hyper-parameter	Value
N estimators	100
Max samples	256
Contamination	0.009
Random state	42
Max features	1
Bootstrap	False

sub-sampling without requiring costly distance or density calculations and scales efficiently to high-dimensional datasets [60].

In this model, anomalies are defined as observations that deviate significantly from the normal operating patterns learned during training. The training dataset consists of historical EAF data without labels; therefore, supervised classification models cannot be applied. Anomalies are not defined through fixed operational limits. Instead, the model identifies observations that are statistically infrequent with respect to the learned representation of normal behavior. Consequently, some detected anomalies may fall within nominal operating limits if their temporal or structural pattern is atypical. The anomaly decision threshold is determined through the contamination parameter of the Isolation Forest, which represents the expected proportion of anomalies. The selected value (0.009) was established via sensitivity analysis over the range [0.001, 0.02]. Higher contamination values increased false positives without improving the detection of disturbances, whereas lower values reduced models sensitivity.

Table 5 summarizes the main hyperparameters, optimized using Optuna, an automated hyperparameter optimization framework. The configuration balances sensitivity and computational efficiency: the number of estimators captures anomalous patterns without excessive cost, the maximum sample size ensures compact representative trees, and the contamination parameter reflects the low expected anomaly rate, minimizing false alarms. Setting the maximum features to one increases partition randomness and anomaly isolation, while bootstrap sampling is disabled to avoid duplicate samples.

Fig. 5 shows the temperature and oxygen concentration time series during EAF operation, with anomalies detected by the IFA algorithm. Temperature anomalies occur mainly at distribution extremes, while oxygen anomalies cluster around sharp spikes. Potential false positives are limited to a small number of observations near the boundaries of normal operating ranges. Potential false negatives may correspond to minor fluctuations within the range of normal variability; however, no visually significant extreme anomalies appear to remain unflagged. Overall, the observed behavior suggests a low false alarm rate while maintaining sensitivity to abrupt and structurally atypical patterns.

Table 6
NN configuration for predicting electric consumption and chemical composition, together with the results obtained from the 5-fold cross-validation procedure. ReLU: Rectified Linear Unit; MSE: Mean Squared Error; MAE: Mean Absolute Error.

Description	NN 1	NN 2
Neurons in the first hidden layer	64	128
Neurons in the second hidden layer	32	64
Neurons in the output layer	1	9
Activation function in hidden layers	ReLU	ReLU
Activation function in the output layer	Linear	Linear
Normalization applied to hidden layers	No	Yes
Optimizer used for training	Adam	Adam
Loss function	MSE	Weighted loss
Metric used for evaluation	MAE	MAE
Learning rate	0.001	0.001
Batch size	32	30
Epochs	500	150
Validation	NN 1	NN 2
Mean MAE	0.29	0.012
Standard deviation	0.016	0.001

5.2.2. Predictive model

Predictive models analyze historical and current data to anticipate future events. The integration of supervised learning models enhances decision-making by identifying complex patterns and relationships in operational data, allowing for accurate predictions of system behavior.

In this work, two predictive models have been developed. The first estimates the EAF power consumption using the loads of 46 distinct charge materials as input features. The second model predicts the chemical composition of the casting, consisting of 9 chemical elements, based on the materials introduced into the EAF before the refining phase. Accurate electricity consumption forecasting supports optimized scheduling by aligning energy-intensive castings with lower electricity price periods. Accurate determination of the molten bath composition before refining ensures product quality, reduces impurities and energy consumption, and supports efficient decarburization [61], while enabling optimized additive selection and operational planning. Fig. 6 illustrates the impact of the amount of different loaded materials on power consumption across six samples. Each sample maintains a constant total load, with varying proportions of individual materials.

Both models have been implemented using a feed-forward NN architecture. This technique was selected for its simplicity, reliable predictive performance, and compatibility with other simulation methods [62]. Table 6 summarizes the configuration of the proposed NNs. The reported results correspond to the mean and standard deviation obtained from a 5-fold cross-validation procedure, in which the model was trained on 80% of the data and validated on the remaining 20%, with systematic rotation of the validation folds to ensure robust performance estimation.

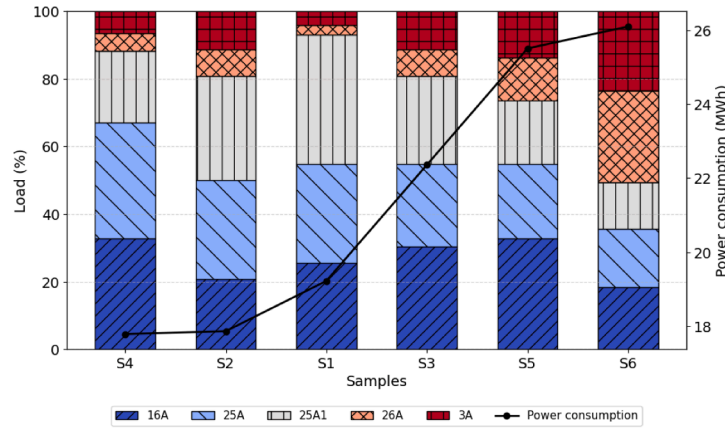


Fig. 6. Power consumption across six samples with the same total load, showing how the quantity of different materials (16A, 25A, 25A1, 26A and 3A) affects energy consumption.

Table 7

Configuration of the XGBoost model and performance results obtained from the 5-fold cross-validation procedure.

Description	Value
N° estimators	142
Max depth	2
Max features	0.77
Min samples split	10
Min samples leaf	2
Metric used for evaluation	MAE
Bootstrap	True
Random state	52
Validation	Value
Mean MAE	466.37
Standard deviation	21.19

Table 8

hyperparameters of the GA and PSO Algorithm.

GA Hyper-parameters	Value
Parent population size	20
Number of generations	6
Number of offspring generated	20
Mutation probability	0.3
Mutation range for K_1	[0, 20000]
Mutation range for $K_{2,3}$	[0, 8000]
Mutation range for $O_{e,r}$	[5, 60]
Mutation range for stop	[60, 6960]
PSO Hyper-parameters	Value
N° of particles	20
N° of iterations	6
Inertia weight	0.7
Learning coefficients	2.1
Initial velocity	Random in [0, 1]

5.2.3. Probability of failure model

Failure probability models are essential for anticipating mechanical degradation and supporting informed operational decisions. In the CCU, ensuring the reliability of sleeves exposed to extreme thermal and mechanical stress is critical to reducing failure risk and minimizing defective production. To address this challenge, an eXtreme Gradient Boosting (XGBoost) model has been developed, incorporating enhanced regularization to prevent overfitting while maintaining high predictive accuracy [63]. The model also identifies the most influential variables affecting sleeve strength. These predictions are used in the decision-making module to optimally schedule sleeve replacement and allocate heats according to the estimated remaining resistance.

The model relies on key input parameters, including the chemical composition of the cast material, the geometry of the piece, and the number of crystallizers, to accurately estimate sleeve resistance. Fig. 7 presents the correlation matrix among the input variables used by the model. Given the large number of features, this analysis helps identify strong relationships and potential redundancies that may affect interpretability and generalization. Correlation values range from -1 to 1, where values above ± 0.8 suggest high dependency. As a result, it enables more effective control of the casting process, improves operational efficiency, reduces defects, and increases the overall quality of the final product. Table 7 presents the configuration of this model, along with the performance results obtained using a 5-fold cross-validation procedure.

To enhance clarity regarding the integration of data-driven and physics-based approaches in plant simulation, Fig. 8 presents the models implemented at each stage of the process, along with their interactions and links to subsequent stages for output variable estimation.

5.3. Decision-making model

A decision-making model is a structured representation that integrates data, variables, and constraints to select the best possible action, improving operational efficiency. In this case two alternative approaches are proposed to optimize the decision-making process in response to disturbances. The first approach optimizes decision-making in near real-time using metaheuristic algorithms, while the second identifies the best solution in real-time, using pre-optimized scenarios and a supervised learning model. Metaheuristic methods are employed because the OF is non-differentiable. The objective is to compare the response quality of the first approach, which sacrifices time, with the speed of the second approach, which compromises response quality.

5.3.1. Metaheuristic algorithms

These algorithms were selected for their simplicity, ease of parallelization, and proven effectiveness in optimizing complex problems [64]. Specifically, GA and PSO were chosen as they represent distinct metaheuristic approaches. GA is an evolutionary algorithm while PSO is based on models of collective behavior. This comparison allows for the identification of approaches that enhance the robustness of the optimization process across different search space structures. The hyperparameters used in the metaheuristic algorithms, GA and PSO, are crucial for improving performance and enabling efficient search in the solution space, ensuring convergence to optimal solutions. Table 8 shows the configuration of these models.

The selection of these hyperparameters is based on balancing exploration. In both methods, a population size of 20 and 6 iterations

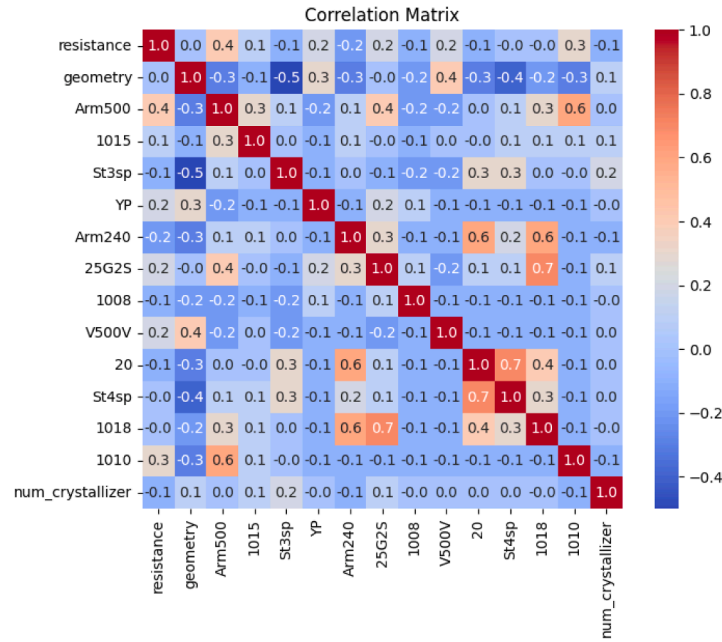


Fig. 7. Correlation matrix of the input variables used in the XGBoost model for predicting sleeve strength.

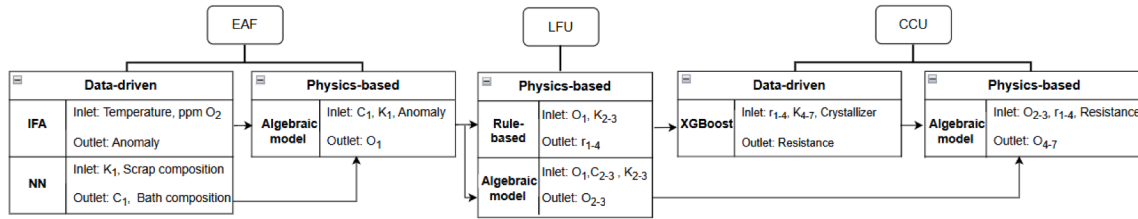


Fig. 8. Schematic representation of the process models and their connections.

were set to ensure computational time remains under 25 minutes while achieving good convergence. In GA, selecting 20 individuals per generation preserves high-quality solutions while introducing new variations. A mutation probability of 0.3 maintains diversity and helps escape local optima. These values were chosen to optimize convergence speed, solution quality, and computational efficiency. In PSO, an inertia weight of 0.7 ensures a balance between global exploration and local refinement, while learning coefficients of 2.1 promote personal and global influence on particle movement. Randomly initialized velocities enhance variability, preventing premature convergence and ensuring adaptive behavior throughout the optimization process.

5.3.2. Supervised learning model

Unlike the previous models, this approach has not performed direct optimization; instead, it has enabled real-time selection of the most suitable solution from a set of pre-optimized scenarios. In this case, a Random Forest (RF) model trained on GA pre-optimized scenarios (GA + RF) has been employed. The dataset has been generated by sampling a wide range of disturbance types and magnitudes, including fluctuations in electricity prices, variations in equipment electric consumption, changes in product demand, equipment failures, and their combined effects. The sampling ranges have been deliberately selected to cover the expected operating envelope of the plant and typical disturbance intensities, ensuring broad representation of relevant perturbations. The resulting dataset has been randomly divided into training (80%) and validation (20%) subsets. The validation set has been excluded from the training phase to assess the generalization capability of the proposed GA + RF approach.

Table 9

Hyper-parameters used for training the RF.

RF Hyper-parameters	
N° estimators	100
Max depth	9
Min samples split	2
Min samples leaf	1

The RF model has been selected for its robustness, as it reduces overfitting and maintains high predictive performance through injected randomness, resulting in competitive results in classification and regression tasks [65]. As RF performance strongly depends on hyperparameter selection, the search ranges have been defined according to the recommendations of Breiman [65] and subsequently optimized using Optuna, balancing model complexity, generalization capability, and computational cost. Table 9 presents the final optimized configuration.

A potential risk of inheriting suboptimal GA solutions was mitigated by performing multiple runs with different initializations and retaining the best-performing solution for each scenario. The GA was configured with population diversity preservation mechanisms, including mutation and crossover strategies, as well as multiple independent runs per scenario. This setup reduces the likelihood of systematic convergence to local optima. The best-performing solution across all runs was retained and added to the database. Furthermore, the RF model does not interpolate individual GA solutions; instead, it learns a correlation between the disturbance in a scenario and the corresponding indices of optimal operational solutions. Consequently, occasional suboptimal GA solutions do

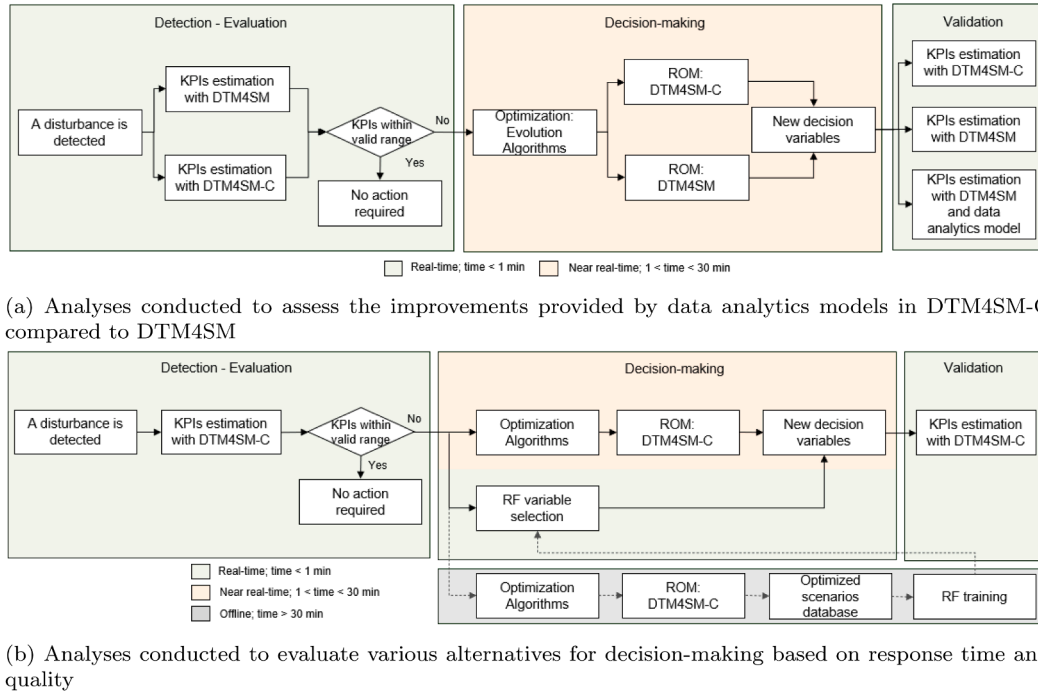


Fig. 9. Analysis conducted to validate DTM4SM-C.

not dominate the learning process, particularly given the ensemble nature of RF and the diversity of scenarios in the training database, which comprises 5000 cases. Finally, the evaluation of generalization using previously unseen combinations of disturbances provides an empirical assessment of the quality and robustness of the pre-optimized solutions. If local optima were prevalent, a degradation in validation performance would be expected. The absence of such degradation suggests that the impact of potential local-optimal convergence is limited.

6. Results

This section presents the analyses conducted to validate the proposed framework and to quantify its improvements over DTM4SM. The first part of this section reports the results of Analysis (a), which evaluates the impact of the newly incorporated models on KPI estimation and on the search for improved operational strategies under different disturbance scenarios. The objective of this analysis is to assess the contribution of the additional modeling components introduced in DTM4SM-C. To ensure a fair and consistent comparison, the solutions generated by DTM4SM are validated within the DTM4SM-C evaluation framework. In addition, the results are benchmarked against a static deterministic baseline, which computes a single offline schedule using a physics-based ROM with nominal parameter values and is executed without reoptimization under disturbed conditions. This additional comparison shows the limitations of static scheduling under disturbances and provides a clear reference for assessing the benefits of dynamic and proactive decision-making strategies. In the second part, the results of analysis (b) are presented, where two optimization techniques close to real-time are compared with the solution proposed in DTM4SM that allows a real-time solution search. A comparison of the time versus quality of the response of each technique to various disturbances is included.

Fig. 9 shows these analyses. In both analyses, the detection of a disturbance in the system acts as the trigger for the evaluation process. The first step consists of assessing the impact of the disturbance on the selected KPIs.

In analysis (a), this assessment is carried out using DTM4SM and DTM4SM-C to enable a direct comparison between the two frameworks. When the KPIs fall outside acceptable ranges, an optimization of the de-

cision variables is activated. This optimization is performed using the corresponding ROMs of DTM4SM and DTM4SM-C. Subsequently, the resulting KPIs obtained from the optimization solutions are evaluated and compared for each ROM. In addition, a validation procedure is included in which the solution generated by DTM4SM is evaluated using predictions provided by data analytics models of DTM4SM-C. This validation addresses an inherent limitation of the DTM4SM approach, which, due to the lack of predictive models, generates solutions without considering anticipated system behavior, potentially leading to unrealistic outcomes. This results in a more representative approximation of real plant conditions and a more accurate evaluation of its effectiveness.

In analysis (b), following disturbance detection, the DTM4SM-C ROM is used to analyze system behavior. Decision-making is evaluated under two alternative strategies. The first relies on a more accurate optimization scheme, adapted to near-real-time plant conditions through the application of metaheuristic algorithms. The second strategy follows the DTM4SM approach, which combines a database of optimized scenarios with an RF to accelerate the decision-making process and enable real-time responses. This model is retrained offline whenever a new disturbance or an unobserved scenario is detected, allowing for progressive performance improvement. Finally, the results obtained from both optimization strategies are compared in terms of solution quality and execution time under different disturbance scenarios, thereby determining which approach offers greater efficiency and robustness in decision-making.

6.1. Analyses conducted to assess the improvements provided by data analytics models in DTM4SM-C compared to DTM4SM

The main differences between the two models compared in the analysis (a) are that in DTM4SM, a linear degradation of the crystallizer sleeve is assuming, whereas in DTM4SM-C, sleeve degradation is estimated using the XGBoost model trained on real plant data. Additionally, in the first case, power consumption follows a linear relationship with the load and process duration, without considering the chemical composition of the mixture charged to the EAF, while in the second case, it depends on the chemical composition of the charged materials, determined through NN models. Furthermore, in DTM4SM-C the duration of the processes

Table 10

Results of KPIs and decision variables under disturbances and after optimization for DTM4SM and DTM4SM-C. In each case, the best OF value is highlighted in bold.

ROM	Case	OF	Δ (%)	CpT	Δ (%)	CO ₂	DemandC	PpM	K ₁	K _{2,3}	O ₁	O _{2,3}	O ₄₋₇
Physics-based	Deferred optimization	7.55	3.28	418.40	6.69	2.23	100.42	20.08	43.46	13.09	44.80	16.61	101.42
DTM4SM	Deferred optimization	7.43	1.64	422.14	7.64	2.25	100.21	20.79	50.10	16.32	51.46	15.35	116.91
DTM4SM-C	Deferred optimization	7.31		392.18		2.35	99.84	23.51	48.63	24.79	49.98	11.77	113.46
Physics-based (Static)	Disturbance A	7.92	4.21	457.96	4.88	2.36	101.46	22.83	48.63	24.79	49.98	11.77	113.46
	Disturbance B	10.22	14.06	448.88	2.94	3.05	101.27	20.59	48.63	24.79	49.98	11.77	113.46
	Disturbance C	10.31	19.47	463.61	14.97	3.05	101.27	20.59	48.63	24.79	49.98	11.77	113.46
DTM4SM (Dynamic)	Disturbance A	7.80	2.63	467.74	7.12	2.35	99.84	23.51	48.63	24.79	49.98	11.77	113.46
	Disturbance A opt.	7.68	4.92	439.39	8.42	2.42	100.32	21.02	31.51	13.88	32.82	7.63	73.53
	Disturbance B	9.53	6.36	480.24	10.13	3.16	100.17	21.82	31.51	13.88	32.82	7.63	73.53
	Disturbance B opt.	8.99	5.64	468.30	17.89	3.04	100.28	22.56	50.51	18.61	51.86	10.19	117.86
	Disturbance C	9.12	5.68	486.76	17.25	3.04	100.28	22.56	50.51	18.61	51.86	10.19	117.86
	Disturbance C opt.	9.01	5.13	487.01	21.20	3.01	99.86	22.62	56.92	19.76	58.29	11.48	132.81
	Temperature anomaly	9.11	5.68	499.71	23.93	3.00	99.86	22.63	56.92	19.76	68.29	11.48	132.81
DTM4SM-C (Proactive)	Disturbance A	7.60		436.66		2.35	99.84	23.51	48.63	24.79	49.98	11.77	113.46
	Disturbance A opt.	7.32		405.26		2.39	99.76	21.32	40.17	21.11	41.38	9.72	93.74
	Disturbance B	8.96		436.05		3.09	99.69	22.15	40.17	21.11	41.38	9.72	93.74
	Disturbance B opt.	8.51		397.23		3.05	99.80	22.48	46.63	24.79	41.29	9.40	108.80
	Disturbance C	8.63		415.16		3.05	99.80	22.48	46.63	24.79	41.29	9.40	108.80
	Disturbance C opt.	8.57		401.81		3.07	99.76	22.24	47.23	24.79	48.27	9.52	110.20
	Temperature anomaly	8.62		403.23		3.07	99.76	22.24	47.23	24.79	48.28	9.52	110.20
DTM4SM with data analysis	Disturbance A	8.04	9.84	500.09	23.40	2.42	99.45	21.06	31.51	13.88	32.82	7.63	73.53
	Disturbance B	9.57	12.46	535.45	34.80	3.04	99.61	22.42	50.51	18.61	51.86	10.19	117.86
	Disturbance C	9.86	15.05	570.22	41.91	3.01	99.18	22.47	56.92	19.76	58.29	11.48	132.81

and the electric consumption are influenced by the anomaly detection models.

Both ROM models have been analyzed under the influence of an unexpected increase of 50% in electricity prices during the period between minute 2880 and 4320 (denoted as disturbance in A), an increase in demand of 30% in minute 4500 (disturbance in B) and an increase of 50% in the power consumption of the CCU unit from minute 6000 (disturbance in C). These disturbances are representative of realistic operational fluctuations commonly observed in steel plants. Increases in electricity prices may result from market volatility, sudden demand surges are often linked to unexpected orders, and variations in energy consumption are typically caused by equipment deterioration or malfunction. The magnitude of these disturbances are based on [47,58] and [48].

Table 10 presents KPI values before and after disturbances, as well as after GA-based optimization, including the corresponding decision variables. It also evaluates a gradual temperature deviation in the EAF (from minute 7000) using each ROMs. The lower section reports results for the third scenario, where the solution given by DTM4SM is evaluated in DTM4SM-C for a fairer comparison. Additionally, a comparison with purely physical and static approaches reported in the Section 2 is included to show the advantages of the proposed proactive framework. Additionally, It reports the relative deviation (%) of the KPI with the highest impact, CpT, and the OF with respect to DTM4SM-C across all comparative scenarios. These deviations serve to quantify structural differences between the ROMs and provide a rigorous assessment of their sensitivity to varying operating conditions.

Fig. 10 provides a graphical comparison of the operation of the EAF, LFU, and CCU units following the implementation of the optimized operating strategy with each ROM for each disturbance within an RHP. The figure illustrates the operating times of each unit, as well as maintenance and production downtime. Subfigures (a) and (b) show the deferred optimization results, depicting the baseline schedule for a full week without disturbances. Subfigures (c) and (d), (e) and (f) and (g) and (h) correspond to the system response to disturbances A, B, and C, respectively. In each case, the disturbance alters the initial schedule, and the optimizer dynamically readjusts load assignments, maintenance start times, and operation durations. Each subfigure also illus-

trates the number of loads processed by each unit and the timing of their operations.

A combined analysis of Table 10 and Fig. 10 highlights how differences between the ROMs affect KPI estimation and operational optimization. Overall, DTM4SM-C delivers the best performance. In the deferred optimization, DTM4SM achieves a CpT that is 7.64% higher relative to DTM4SM-C and 6.69% higher compared to the physics-based ROM. These deviations show the structural performance differences between the models and underscore the benefit of predictive analytics and proactive anticipation of future operating conditions in the deferred optimization stage. Furthermore, DTM4SM-C enables a more efficient scheduling strategy of operation and maintenance. DTM4SM-C schedules the CCU maintenance at minute 820, while DTM4SM delays it to minute 2138, due to its consideration of both load and material composition for predicting wear.

For disturbance A, both models advance maintenance and production downtime to periods with higher electricity costs and reduce EAF load to lower energy consumption. However, DTM4SM-C achieves a smaller reduction by alternating charged materials, prioritizing those requiring less energy, and ensures production finishes before Saturday, yielding a 4.69% improvement in OF. Disturbance B negatively impacts KPIs. While DTM4SM achieves higher CO₂ reduction, DTM4SM-C attains lower cost and better demand fulfillment. Both models increase equipment loads, but DTM4SM-C slightly reduces operation duration, minimizing weekend work and lowering OF by 5.34%. For disturbance C, DTM4SM prioritizes emissions reduction and demand alignment by increasing loads and process durations, reducing Sunday work. DTM4SM-C, however, focuses on reducing EAF and CCU loads and overall costs, achieving a 4.88% OF improvement. EAF temperature deviations above operational limits require early adjustments. DTM4SM detects anomalies only when alarm thresholds are reached, resulting in a 1.11% OF deterioration; however, DTM4SM-C incorporates an IFA to identify anomalies at earlier stages, enabling smoother corrective actions and reducing the OF impact to 0.58%.

A third experimental scenario evaluates DTM4SM solutions in an environment with predictive information. Under these conditions, DTM4SM-C achieves OF improvements of 9.84%, 11.08%, and 13.08% for disturbances A, B, and C, respectively, corresponding to an average CpT reduction of 133.82 \$/t in CpT. These results demonstrate the

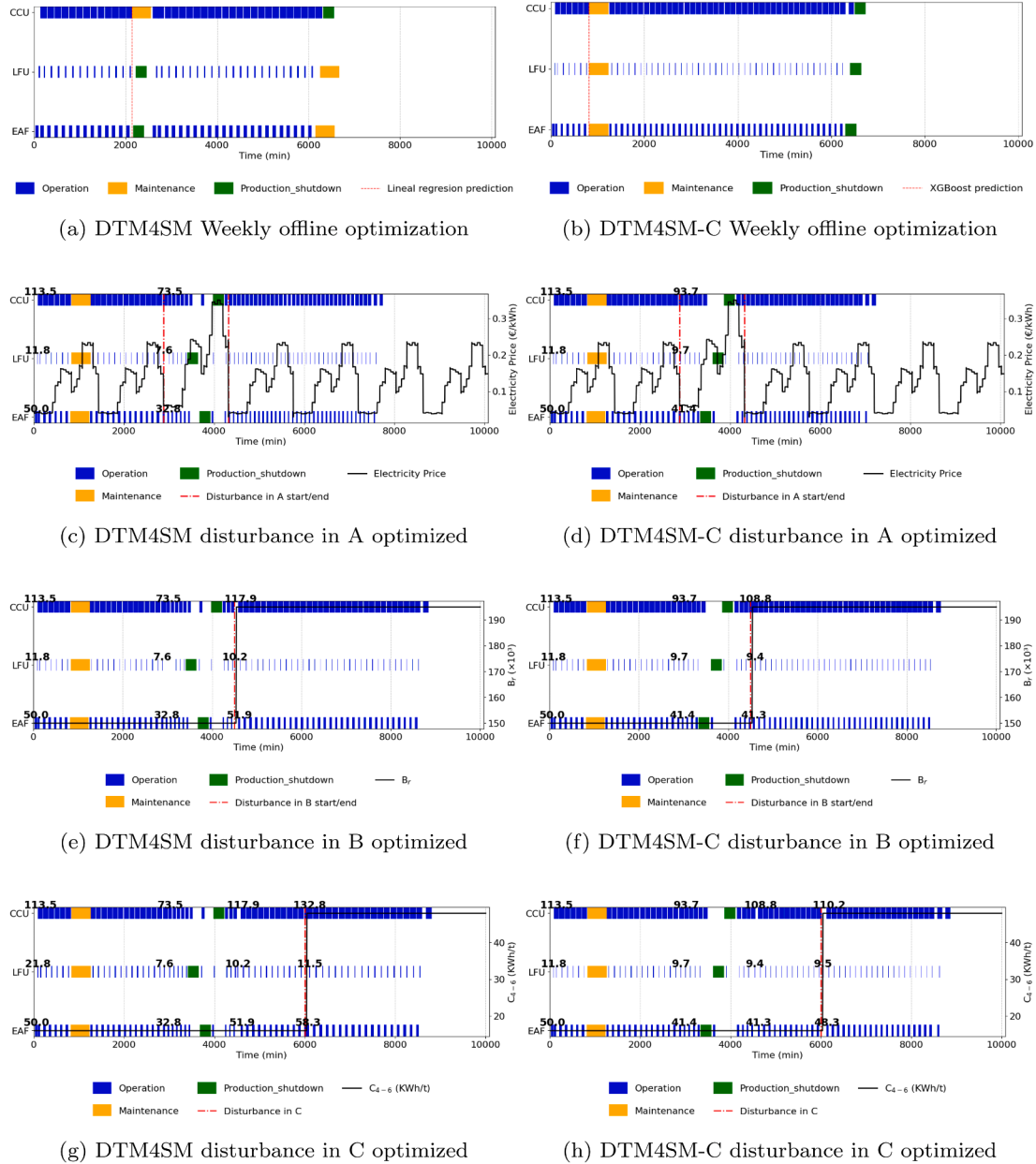


Fig. 10. Optimization results over a RHP using ROM-based methods DTM4SM and DTM4SM-C, showing the distribution of production and maintenance shutdown, as well as operating periods, across the three main processes under multiple disturbances.

effectiveness of DTM4SM-C in enabling early anomaly detection and mitigating disturbances. The framework dynamically adjusts decision variables to optimize KPIs and ensure optimal plant operation under changing conditions.

6.2. Analyses conducted to evaluate different alternatives for decision-making based on response time and quality

The decision-making models are evaluated to assess their ability to provide efficient, robust, and practical solutions in manufacturing scenarios affected by disturbances and operational failures. To this end, different optimization strategies are compared for both DTM4SM-C and DTM4SM, with particular emphasis on their performance in quasi-real-time and real-time decision-making contexts. The analysis includes the metaheuristic optimization approaches, GA and PSO, as well as a hybrid GA + RF strategy designed to deliver high-quality solutions with immediate response times. This hybrid approach relies on an RF model trained

on GA-optimized scenarios, enabling near-instantaneous solutions and making it suitable for time-critical production environments.

Comparative results are reported in Table 11, where GA + RF is evaluated against the other algorithms in terms of solution quality and computational time. Performance improvements are quantified using the OF and the corresponding KPIs. Fig. 11 illustrates the OF comparison across the ten analyzed cases. The robustness of each approach is assessed under distinct scenarios, including variations in electricity prices, product demand, energy consumption, as well as operational failures in critical production units. All optimization experiments were executed on a workstation equipped with an Intel® Xeon® Gold 5217 CPU at 3.00 GHz (8 cores) and 192 GB of RAM, ensuring a consistent experimental environment.

The evaluation considers two optimized weekly schedules, each subjected to a set of disturbances and failures. In the first experiment, disturbances A, B, and C represent changes in electricity prices, product demand, and equipment energy consumption, while failures in the EAF

Table 11

Comparison of optimizer results for different disturbances and breakages. The best OF result for each case is shown in bold.

Optimizer	Case	OF	Δ (%)	CpT	Δ (%)	CO ₂	DemandC	PpM	Water
GA	Perturbación A	7.25		323.94		2.42	99.51	19.04	9.50
	Perturbación B	8.72		242.07		3.06	100.15	20.40	9.46
	Perturbación C	8.60		235.15		3.04	99.76	20.40	9.43
	Fallo EAF	8.77		177.61		3.05	98.97	19.52	9.41
	Fallo LFU	9.36		181.55		3.04	98.91	18.73	9.41
	Perturbación A'	6.21		219.35		2.26	99.44	20.33	9.47
	Perturbación B'	7.78		186.76		2.95	99.94	21.20	9.46
	Perturbación C'	7.79		185.20		2.95	100.16	21.24	9.46
	Fallo EAF'	8.25		189.38		2.93	99.43	21.51	9.43
	Fallo LFU'	8.42		140.51		2.92	99.20	19.63	9.48
PSO	Perturbación A	7.17	-1.10	319.51	-1.37	2.32	98.61	19.31	9.52
	Perturbación B	9.01	3.33	305.25	26.10	3.04	99.34	20.50	9.49
	Perturbación C	8.51	-1.05	195.20	-16.99	3.05	98.97	20.10	9.41
	Fallo EAF	9.04	3.08	212.37	19.57	3.06	99.85	19.57	9.45
	Fallo LFU	9.38	0.21	184.41	1.57	3.04	98.91	18.73	9.41
	Perturbación A'	6.24	0.48	196.55	-10.39	2.25	100.05	19.40	9.49
	Perturbación B'	7.69	-1.16	188.21	0.78	2.92	98.95	21.32	9.47
	Perturbación C'	7.80	0.13	185.18	-0.01	2.95	100.24	21.24	9.46
	Fallo EAF'	8.28	0.36	196.69	3.86	2.93	99.42	20.38	9.42
	Fallo LFU'	8.77	4.16	189.01	34.52	2.93	99.54	19.54	9.42
GA + RF	Perturbación A	7.55	4.14	383.37	18.35	2.35	99.16	19.25	9.46
	Perturbación B	9.11	4.47	288.51	19.18	3.09	100.01	20.28	9.48
	Perturbación C	8.62	0.23	235.70	0.23	3.04	99.87	20.40	9.44
	Fallo EAF	8.87	1.14	186.82	5.19	3.05	99.17	19.50	9.43
	Fallo LFU	9.38	0.21	182.85	0.72	3.06	98.34	18.75	9.43
	Perturbación A'	6.24	0.48	219.87	0.24	2.25	99.09	20.31	9.45
	Perturbación B'	7.83	0.64	213.20	14.16	2.92	99.15	21.35	9.48
	Perturbación C'	7.96	2.18	213.05	15.04	2.95	100.03	21.20	9.46
	Fallo EAF'	8.27	0.24	188.17	-0.64	2.94	99.72	20.35	9.44
	Fallo LFU'	8.67	2.97	188.84	34.49	2.92	99.99	19.79	9.52

and LFU units cause four-hour shutdowns at minutes 7000 and 7500, respectively. In the second experiment, the same disturbance types, denoted as A' , B' , and C' , and failures EAF' and LFU' , are redistributed in time at minutes 3200, 5000, 6500, 7300, and 7760. These scenarios enable a systematic assessment of optimization robustness under different operational conditions and disturbances.

Analyzing the results across the ten evaluated cases, GA achieved the lowest OF values in 7 cases, while PSO obtained the minimum OF in the remaining 3. GA achieved the best OF values in disturbances B, EAF, LFU, A', C', EAF', and LFU', consistently improving CpT, CO₂ emissions, water consumption, and, in some cases, achieving a higher PpM, which helped minimize weekend work penalties. PSO performed best in disturbances A, C and B', mainly reducing CpT, CO₂ emissions, and water consumption.

Although the GA + RF approach did not achieve the minimum OF in any individual case, it demonstrated consistent near-optimal performance across all scenarios while drastically reducing computation time, with an average of 0.27 minutes. In contrast, the metaheuristic approaches required up to 25 minutes, with most OF improvements occurring after approximately 15 minutes. On average across the ten scenarios, GA + RF yields OF values that are 1.62% higher than GA and 0.81% higher than PSO. These differences correspond to mean increases in CpT of 21.89 \$/t and 12.81 \$/t respectively. Reported percentages therefore represent mean relative differences, while maximum improvements are highlighted where explicitly stated. Overall, GA + RF provides a robust and computationally efficient alternative for time-constrained applications where fast, near-optimal decisions are required.

7. Conclusions

In the manufacturing sector, the use of DTs enhances operational efficiency and enables predictive modeling under anomalous conditions. When combined with AI, DTs allow accurate simulation of system dynamics and the real-time generation of optimized and proactive re-

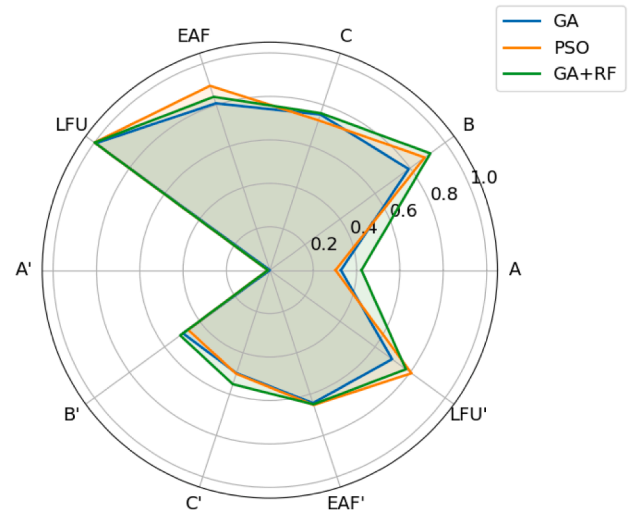


Fig. 11. Comparison of the OF values across the analyzed cases.

sponses to evolving operational and maintenance requirements. However, a significant gap remains in the integration of hybrid models and data analytics at the plant-wide level. Most existing approaches rely primarily on physics-based models and focus on individual assets, which limits their ability to capture overall system behavior and to support robust decision-making under uncertainty.

To address this gap, this paper proposes DTM4SM-C, an enhanced framework designed to support decision-making in unforeseen scenarios using a RHP. The proposed approach integrates anomaly detection, event forecasting, and failure probability estimation, leveraging ROMs trained on both synthetic and real data to capture complex and hard-to-model system interactions. These models offer fast execution, mak-

ing them suitable for real-time applications. In addition, DTM4SM-C introduces a multi-criteria optimization framework for operation and maintenance planning, improving adaptability, recommendation reliability, and alignment between production, demand, and economic and environmental objectives. Overall, DTM4SM-C extends DTM4SM from reactive analysis toward proactive and predictive decision-making in complex industrial systems.

DTM4SM-C has been validated using operational data from a real steel plant, considering the influence of various types of disturbances within a defined time horizon, including fluctuations in electricity prices, variations in product demand, and unexpected failures in the EAF and LFU units. The analysis was conducted using the ROMs developed in DTM4SM and DTM4SM-C, enabling an assessment of the dynamic adaptability of both systems under different scenarios. The results show that the DTM4SM-C ROM achieves an average OF improvement of 11.33%, corresponding to an average CpT reduction of 133.82\$/t.

Regarding the optimized decision-making model, significant performance improvements are observed with the use of metaheuristic algorithms. The comparative assessment over ten disturbed operating scenarios indicates that, although GA achieves the lowest OF values in most cases, GA + RF approach consistently delivers near-optimal solutions, with average OF deviations below 1.62% and a substantial reduction in computation time from several minutes to seconds, thus representing a viable solution for real-time decision-making in industrial environments subject to strict time constraints.

The framework is designed with a modular architecture that allows practitioners to configure and adapt it to the specific needs of their plants. Users select the models within each module according to process characteristics, data availability, and operational constraints, enabling application across different facilities, data volumes, and computational capacities. This modularity also supports cross-cutting tasks such as dynamic rolling-horizon optimization under disturbances.

Operationally, the framework supports real-time adjustment of operation and maintenance strategies through continuous data analysis, KPIs estimation, and scenario evaluation. By providing optimized strategies based on economic and environmental KPIs, it enables efficient resource management, prioritized interventions, and process stability under unforeseen variations, making it applicable to sectors such as manufacturing, logistics, and energy.

Future work will be directed towards the implementation of DTM4SM-C within an industrial environment. A pilot study will be conducted on a production line to enable a controlled and systematic evaluation. The pilot phase will include real-time data acquisition, integration with existing control systems, and testing under diverse conditions to evaluate the effectiveness, robustness, and adaptability of DTM4SM-C to disturbances. Based on the findings, an evaluation will be conducted to assess its potential scalability and ensure a structured transition to full-scale implementation.

CRedit authorship contribution statement

Rocío Mingorance Mingorance: Writing - original draft, Validation, Software, Methodology, Investigation, Data curation, Conceptualization; **Diego Crespo Pereira:** Writing - review & editing, Supervision, Conceptualization; **Jone Uribetxebarria:** Writing - review & editing, Validation, Supervision, Investigation, Conceptualization; **Urko Leturiondo:** Writing - review & editing, Supervision.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Rocio Mingorance reports financial support was provided by Basque Government. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This study was partially funded by the MECACOGNIT project, economically supported by The Basque Government under Project No.Exp KK-2024/00030 in the ELKARTEK program.

References

- [1] R.M. Yazdi, F. Imani, H. Yang, A hybrid deep learning model of process-build interactions in additive manufacturing, *J. Manuf. Syst.* 57 (2020) 460–468. <https://doi.org/10.1016/j.jmsy.2020.11.001>
- [2] Y. Gao, D. Chang, C.-H. Chen, M. Sha, A digital twin-Based decision support approach for AGV scheduling, *Eng. Appl. Artif. Intell.* 130 (2024) 107687. <https://doi.org/10.1016/j.engappai.2023.107687>
- [3] D. Pagano, A predictive maintenance model using long short-Term memory neural networks and bayesian inference, *Decision Anal. J.* 6 (2023) 100174. <https://doi.org/10.1016/j.dajour.2023.100174>
- [4] M. Yuan, Z. Li, C. Zhang, L. Zheng, K. Mao, F. Pei, Research on real-time prediction of completion time based on AE-CNN-LSTM, *Comput. Ind. Eng.* 185 (2023) 109677. <https://doi.org/10.1016/j.cie.2023.109677>
- [5] L. Wang, B. Li, X. Zhao, Multi-objective predictive maintenance scheduling models integrating remaining useful life prediction and maintenance decisions, *Comput. Ind. Eng.* 197 (2024) 110581. <https://doi.org/10.1016/j.cie.2024.110581>
- [6] J. Li, Z. Wang, S. Zhang, Y. Lin, L. Jiang, J. Tan, Task incremental learning-Driven digital-Twin predictive modeling for customized metal forming product manufacturing process, *Robot Comput. Integr. Manuf.* 85 (2024) 102647. <https://doi.org/10.1016/j.rcim.2023.102647>
- [7] Y. Lu, C. Liu, K.I.-K. Wang, H. Huang, X. Xu, Digital twin-driven smart manufacturing: connotation, reference model, applications and research issues, *Robot Comput. Integr. Manuf.* 61 (2020) 101837. <https://doi.org/10.1016/j.rcim.2019.101837>
- [8] A. Mukherjee, A.B. Divya, M. Sivvani, S.K. Pal, Cognitive intelligence in industrial robots and manufacturing, *Comput. Ind. Eng.* 191 (2024) 110106. <https://doi.org/10.1016/j.cie.2024.110106>
- [9] G. Gao, D. Zhou, H. Tang, X. Hu, An intelligent health diagnosis and maintenance decision-making approach in smart manufacturing, *Reliab. Eng. Syst. Saf.* 216 (2021) 107965. <https://doi.org/10.1016/j.res.2021.107965>
- [10] O. Wu, G. Dalle Ave, I. Harjunkoski, L. Imsland, A rolling horizon approach for scheduling of multiproduct batch production and maintenance using generalized disjunctive programming models, *Comput. Chemical Eng.* 148 (2021) 107268. <https://doi.org/10.1016/j.compchemeng.2021.107268>
- [11] S. Abedi, S. Kwon, Rolling-horizon optimization integrated with recurrent neural network-driven forecasting for residential battery energy storage operations, *Int. J. Electrical Power & Energy Syst.* 145 (2023) 108589. <https://doi.org/10.1016/j.ijepes.2022.108589>
- [12] M. Klar, P. Ruediger, M. Schuermann, G.T. Gören, M. Glatt, B. Ravani, J.C. Aurich, Explainable generative design in manufacturing for reinforcement learning based factory layout planning, *J. Manuf. Syst.* 72 (2024) 74–92. <https://doi.org/10.1016/j.jmsy.2023.11.012>
- [13] D. Ivanov, Conceptualisation of a 7-element digital twin framework in supply chain and operations management, *Int. J. Prod. Res.* 62 (6) (2024) 2220–2232. <https://doi.org/10.1080/00207543.2023.2217291>
- [14] H. Li, D. Yang, H. Cao, W. Ge, E. Chen, X. Wen, C. Li, Data-driven hybrid petri-net based energy consumption behaviour modelling for digital twin of energy-efficient manufacturing system, *Energy* 239 (2022) 122178. <https://doi.org/10.1016/j.energy.2021.122178>
- [15] O. Kaikova, V. Terziyan, Deep neural networks, cellular automata and petri nets: useful hybrids for smart manufacturing, *Procedia Comput. Sci.* 232 (2024) 2334–2346. 5th International Conference on Industry 4.0 and Smart Manufacturing (ISM 2023). <https://doi.org/10.1016/j.procs.2024.02.052>
- [16] R. Mingorance, D.C. Pereira, J. Uribetxebarria, U. Leturiondo, A methodology leveraging digital twins to enhance the operational strategy of manufacturing plants in unexpected scenarios, *Results Eng.* 27 (2025) 106761. <https://doi.org/10.1016/j.rineng.2025.106761>
- [17] S.-L. Jiang, C. Xu, L. Zhang, Y. Ma, A decomposition-based two-stage online scheduling approach and its integrated system in the hybrid flow shop of steel industry, *Expert Syst. Appl.* 213 (2023) 119200. <https://doi.org/10.1016/j.eswa.2022.119200>
- [18] J. Long, Z. Sun, P.M. Pardalos, Y. Bai, S. Zhang, C. Li, A robust dynamic scheduling approach based on release time series forecasting for the steelmaking-continuous casting production, *Appl. Soft. Comput.* 92 (2020) 106271. <https://doi.org/10.1016/j.asoc.2020.106271>
- [19] Y. Zhou, H. Xiang, W. Zhou, W. Liu, R. Zhang, Proactive scheduling of steelmaking-continuous casting with uncertain processing times under carbon emission reduction, *Chem. Eng. Res. Des.* 212 (2024) 421–433. <https://doi.org/10.1016/j.cherd.2024.11.023>
- [20] M.P. Fanti, G. Rotunno, G. Stecco, W. Ukovich, S. Mininel, An integrated system for production scheduling in steelmaking and casting plants, *IEEE Trans. Autom. Sci. Eng.* 13 (2) (2016) 1112–1128. <https://doi.org/10.1109/TASE.2015.2477362>

- [21] Y. Tan, S. Liu, Models and optimisation approaches for scheduling steelmaking-refining-continuous casting production under variable electricity price, *Int. J. Prod. Res.* 52 (4) (2014) 1032–1049. <https://doi.org/10.1080/00207543.2013.828179>
- [22] D.-L. Hou, T.-K. Li, Analysis of random disturbances on shop floor in modern steel production dynamic environment, *Procedia Eng.* 29 (2012) 663–667. 2012 International Workshop on Information and Electronics Engineering. <https://doi.org/10.1016/j.proeng.2012.01.020>
- [23] J.-Q. Li, Q.-K. Pan, K. Mao, A hybrid fruit fly optimization algorithm for the realistic hybrid flowshop rescheduling problem in steelmaking systems, *IEEE Trans. Autom. Sci. Eng.* 13 (2) (2016) 932–949. <https://doi.org/10.1109/TASE.2015.2425404>
- [24] Y. Lv, B. Qian, R. Hu, H.-P. Jin, Z.-Q. Zhang, An enhanced cross-entropy algorithm for the green scheduling problem of steelmaking and continuous casting with uncertain processing time, *Comput. Ind. Eng.* 171 (2022) 108445. <https://doi.org/10.1016/j.cie.2022.108445>
- [25] L. Tang, Y. Zhao, J. Liu, An improved differential evolution algorithm for practical dynamic scheduling in steelmaking-Continuous casting production, *IEEE Trans. Evol. Comput.* 18 (2) (2014) 209–225. <https://doi.org/10.1109/TEVC.2013.2250977>
- [26] Q.-K. Pan, An effective co-evolutionary artificial bee colony algorithm for steelmaking-continuous casting scheduling, *Eur. J. Oper. Res.* 250 (3) (2016) 702–714. <https://doi.org/10.1016/j.ejor.2015.10.007>
- [27] M.H. Fazel Zarandi, F. Dorry, A hybrid fuzzy PSO algorithm for solving steelmaking-continuous casting scheduling problem, *Int. J. Fuzzy Syst.* 20 (1) (2018) 219–235. <https://doi.org/10.1007/s40815-017-0331-0>
- [28] Z. Xu, Z. Zheng, X. Gao, Energy-efficient steelmaking-continuous casting scheduling problem with temperature constraints and its solution using a multi-objective hybrid genetic algorithm with local search, *Appl. Soft Comput.* 95 (2020) 106554. <https://doi.org/10.1016/j.asoc.2020.106554>
- [29] X. Lian, Z. Zheng, C. Wang, X. Gao, An energy-efficient hybrid flow shop scheduling problem in steelmaking plants, *Computers & Industrial Engineering* 162 (2021) 107683. <https://doi.org/10.1016/j.cie.2021.107683>
- [30] V. Manojlović, Ž. Kamberović, M. Korać, M. Dotlić, Machine learning analysis of electric arc furnace process for the evaluation of energy efficiency parameters, *Appl. Energy* 307 (2022) 118209. <https://doi.org/10.1016/j.apenergy.2021.118209>
- [31] G. Andonovski, S. Tomažič, Comparison of data-Based models for prediction and optimization of energy consumption in electric arc furnace (EAF), *IFAC-PapersOnLine* 55 (20) (2022) 373–378. 10th Vienna International Conference on Mathematical Modelling MATHMOD 2022. <https://doi.org/10.1016/j.ifacol.2022.09.123>
- [32] Y. Saboohi, A. Fathi, I. Škrjanc, V. Logar, Optimization of the electric arc furnace process, *IEEE Trans. Ind. Electron.* 66 (10) (2019) 8030–8039. <https://doi.org/10.1109/TIE.2018.2883247>
- [33] S. Panagou, F. Fruggiero, C. del Vecchio, K. Sarda, F. Menchetti, L. Piedimonte, O.R. Natale, S. Passariello, Explorative hybrid digital twin framework for predictive maintenance in steel industry, in: *IFAC-PapersOnLine*, 55, 2022, pp. 289–294. <https://doi.org/10.1016/j.ifacol.2023.01.087>
- [34] Y.-Z. Wang, Z. Zheng, S.-Y. Zhang, X.-Q. Gao, A robust optimization method for multi-cast batching plans and casting start time dynamic decision in continuous casting process, *Comput. Ind. Eng.* 197 (2024) 110587. <https://doi.org/10.1016/j.cie.2024.110587>
- [35] Z.Z. Jianyu Long, X. Gao, Dynamic scheduling in steelmaking-continuous casting production for continuous caster breakdown, *Int. J. Prod. Res.* 55 (11) (2017) 3197–3216. <https://doi.org/10.1080/00207543.2016.1268277>
- [36] Y.-H.L. Kung-Jeng Wang, S. Angelica, Digital twin design for real-time monitoring - a case study of die cutting machine, *Int. J. Prod. Res.* 59 (21) (2021) 6471–6485. <https://doi.org/10.1080/00207543.2020.1817999>
- [37] J. Li, X. Wang, Q. Yang, Y. Sun, J. Zhao, X. Mao, H. Qie, IoT-Based framework for digital twins in steel production: a case study of key parameter prediction and optimization for CSR, *Expert Syst. Appl.* 250 (2024) 123909. <https://doi.org/10.1016/j.eswa.2024.123909>
- [38] J. Wang, Q. Wang, W. Sun, Optimal power system flexibility-based scheduling in iron and steel production: a case of steelmaking-refining-continuous casting process, *J. Clean Prod.* 414 (2023) 137619. <https://doi.org/10.1016/j.jclepro.2023.137619>
- [39] S. Yu, T. Chai, Y. Tang, An effective heuristic rescheduling method for steelmaking and continuous casting production process with multirefining modes, *IEEE Trans. Syst. Man Cybern. Syst.* 46 (12) (2016) 1675–1688. <https://doi.org/10.1109/TSMC.2016.2604081>
- [40] S.-L. Jiang, Z. Zheng, M. Liu, A multi-stage dynamic soft scheduling algorithm for the uncertain steelmaking-continuous casting scheduling problem, *Appl. Soft. Comput.* 60 (2017) 722–736. <https://doi.org/10.1016/j.asoc.2017.07.016>
- [41] L. Glomb, F. Liers, F. Rösel, A rolling-horizon approach for multi-period optimization, *Eur. J. Oper. Res.* 300 (1) (2022) 189–206. <https://doi.org/10.1016/j.ejor.2021.07.043>
- [42] W. Tian, Y. Cao, Evaluation model and algorithm optimization of intelligent manufacturing system on the basis of BP neural network, *Intelligent Systems with Applications* 20 (2023) 200–293. <https://doi.org/10.1016/j.iswa.2023.200293>
- [43] L. Jia, J. Liu, A. Chong, X. Dai, Deep learning and physics-based modeling for the optimization of ice-based thermal energy systems in cooling plants, *Appl. Energy* 322 (2022) 119443. <https://doi.org/10.1016/j.apenergy.2022.119443>
- [44] R. Santos, H. Piqueiro, R. Dias, C.D. Rocha, Transitioning trends into action: a simulation-Based digital twin architecture for enhanced strategic and operational decision-Making, *Computers & Industrial Engineering* 198 (2024) 110616. <https://doi.org/10.1016/j.cie.2024.110616>
- [45] P. Leitão, A.W. Colombo, S. Karnouskos, Industrial automation based on cyber-physical systems technologies: prototype implementations and challenges, *Comput. Ind.* 81 (2016) 11–25. Emerging ICT concepts for smart, safe and sustainable industrial systems. <https://doi.org/10.1016/j.compind.2015.08.004>
- [46] J.L.E. Berciano, E.T. Guerra, S. de Elío, D.F. Segovia, *Monografías sobre Tecnología del Acero. Parte I: Acería Eléctrica*, Universidad Politécnica de Madrid, 2009.
- [47] I. Katser, I. Mozolin, M. Mezhev, V. Travnikov, V. Kozitsin, RUL Dataset from Continuous Casting Machine, 2023, Available at: <https://www.kaggle.com/ds/1671405>. Last accessed: June 16, 2024. <https://doi.org/10.34740/KAGGLE/DS/1671405>
- [48] I. Katser, M. Mezhev, I. Mozolin, V. Travnikov, V. Kozitsin, RUL Dataset from Continuous Casting Machine, 2023, Available at: <https://www.kaggle.com/ds/4014009>. Last accessed: June 16, 2024. <https://doi.org/10.34740/KAGGLE/DS/4014009>
- [49] S. Yu, A.C. Garrabrants, R.C. DeLapp, T. Hubner, S.A. Thorne, D.S. Kosson, From leaching data to release estimates: screening and scenario assessments of electric arc furnace (EAF) slag under unencapsulated use, *J. Hazard. Mater.* 479 (2024) 135522. <https://doi.org/10.1016/j.jhazmat.2024.135522>
- [50] M.M. Rashid, P. Mhaskar, C.L.E. Swartz, Multi-rate modeling and economic model predictive control of the electric arc furnace, *J. Process Contr.* 40 (2016) 50–61. <https://doi.org/10.1016/j.jprocont.2015.12.012>
- [51] M.M. Abadi, H. Tang, M.M. Rashidi, A review of simulation and numerical modeling of electric arc furnace (EAF) and its processes, *Heliyon* 10 (11) (2024) e32157. <https://doi.org/10.1016/j.heliyon.2024.e32157>
- [52] S. Louhenkilpi, Chapter 1.10 - Continuous casting of steel, in: S. Seetharaman, R. Guthrie, A. McLean, S. Seetharaman, H.Y. Sohn (Eds.), *Treatise on Process Metallurgy*, Elsevier, 2nd edition, 2024, pp. 343–383. <https://doi.org/10.1016/B978-0-323-85373-6.00009-0>
- [53] J.L.E. Berciano, E.T. Guerra, S. de Elío, D.F. Segovia, *Monografías sobre Tecnología del Acero. Parte II: Colada del Acero*, Universidad Politécnica de Madrid, 2009.
- [54] A. Thelen, X. Zhang, O. Fink, et al., A comprehensive review of digital twin — part 1: modeling and twinning enabling technologies, *Struct. Multidiscip. Optim.* 65 (12) (2022) 354. <https://doi.org/10.1007/s00158-022-03425-4>
- [55] U.S. E. P.A. (USEPA), *Compilation of Air Pollutant Emission Factors, Volume 1: Stationary Point and Area Sources*, fifth edition, AP-42. Section 12.10 Grey Iron Foundries, AP-42 Section 12.10, United States Environmental Protection Agency, 2003.
- [56] P. Suvio, A. Van Hoorn, M. Szabo, Å. Ekdahl, Water management for sustainable steel industry, *Ironmak. Steelmaking* 39 (4) (2012) 263–269. <https://doi.org/10.1179/030192311X13135947813898>
- [57] S. Iron, International and Institute, *Producción de acero: Consenso de expertos y IISI Environmental Performance Indicators 2003*, Technical Report, International Iron and Steel Institute, 2004.
- [58] R.E. de España, Datos de Hoy, 2024, (<https://www.ree.es/es/datos/aldia>). Last accessed: 18-01-2024.
- [59] D. Diakoulaki, G. Mavrotas, L. Papayannakis, Determining objective weights in multiple criteria problems: the CRITIC method, *Comput. Oper. Res.* 22 (7) (1995) 763–770. [https://doi.org/10.1016/0305-0548\(94\)00059-H](https://doi.org/10.1016/0305-0548(94)00059-H)
- [60] F.T. Liu, K.M. Ting, Z.-H. Zhou, Isolation forest, in: *Proceedings of the 2008 Eighth IEEE International Conference on Data Mining (ICDM)*, IEEE, 2009, pp. 413–422. <https://doi.org/10.1109/ICDM.2008.17>
- [61] M.M. Elkoumy, M. El-Anwar, A.M. Fathy, G.M. Megahed, I. El-Mahallawi, H. Ahmed, Simulation of EAF refining stage, *Ain Shams Eng. J.* 9 (4) (2018) 2781–2793. <https://doi.org/10.1016/j.asej.2017.10.002>
- [62] S. Curteanu, M. Smarandoiu, D. Horoba, F. Leon, Naphthalene sublimation. experiment and optimisation based on neuro-evolutionary methodology, *J. Ind. Eng. Chem.* 20 (4) (2014) 1608–1611. <https://doi.org/10.1016/j.jiec.2013.08.007>
- [63] S. Ma, J. Gu, J. Wang, Y. Shao, Z. Zhang, X. Liu, In-depth insight into the driving factors of the compressive strength development of MKPC based on interpretable machine learning methods, *J. Ind. Eng. Chem.* 141 (2025) 305–318. <https://doi.org/10.1016/j.jiec.2024.06.039>
- [64] V. Pandhare, E. Negri, L. Ragazzini, L. Cattaneo, M. Macchi, J. Lee, Digital twin-enabled robust production scheduling for equipment in degraded state, *J. Manuf. Syst.* 74 (2024) 841–857. <https://doi.org/10.1016/j.jmsy.2024.04.027>
- [65] L. Breiman, Random forests, *Mach. Learn.* 45 (1) (2001) 5–32. <https://doi.org/10.1023/A:1010933404324>